CHEMISTRY QUESTION BANK

Answer Companion

500 structured answers · 7 units · original PDF page references

Prepared for **Revision12.pdf**

Source contains 28 searchable pages and questions numbered 1–500. The companion preserves the original unit sequence and links each answer to the source PDF page.

Reading guide

Each entry shows the question number, the source tag as printed, and the original PDF page. The answer follows; question wording is kept in the separate inventory to avoid duplicating the source bank.

The source does not define [M] and [H]; they are retained verbatim and are not interpreted as marks or a specific difficulty scale.

This is a carefully reviewed study aid, not an official CBSE marking scheme. Condition-sensitive questions and limitations are documented in the verification report.

Build date: 2026-10-30

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UNIT 1: SOLUTIONS

Questions 1–72 · Original PDF pages 1–4

Question 1 [M] · Original PDF p. 1

Answer. A solution is a homogeneous mixture. Unsaturated solutions can dissolve more solute at the stated temperature; saturated solutions are at equilibrium with undissolved solute; supersaturated solutions contain more dissolved solute than the equilibrium solubility and are metastable.

Question 2 [M] · Original PDF p. 1

Answer. For a non-volatile solute, only solvent contributes appreciably to vapour. Its mole fraction is below one, so Raoult's law gives $p_{solvent}=x_{solvent}\,p^\circ_{solvent}<p^\circ_{solvent}$.

Question 3 [M] · Original PDF p. 1

Answer. For ideal binary volatile liquids A and B: $p_A=x_{ApA}{}^\circ$, $p_B=x_{BpB}{}^\circ$ and $P_{total}=p_A+p_B=x_{ApA}{}^\circ+x_{BpB}{}^\circ$.

Question 4 [H] · Original PDF p. 1

Answer. For solvent A and non-volatile solute B, $p_A=x_{ApA}{}^\circ$. Thus $(p_A{}^\circ-p_A)/p_A{}^\circ=1-x_A=x_B$, the solute mole fraction. This assumes an ideal dilute solution and negligible solute vapour pressure.

Question 5 [M] · Original PDF p. 1

Answer. An ideal solution obeys Raoult's law at every composition. Mixing has $\Delta H_{mix}=0$ and $\Delta V_{mix}=0$; unlike and like intermolecular interactions are approximately equal.

Question 6 [H] · Original PDF p. 1

Answer. Positive deviation occurs when A–B attractions are weaker than A–A/B–B, raising vapour pressure (e.g. ethanol–acetone approximately). Negative deviation occurs when A–B attractions are stronger, lowering vapour pressure (e.g. acetone–chloroform).

Question 7 [M] · Original PDF p. 1

Answer. Unlike molecules attract one another less strongly than like molecules; they escape more readily to vapour, so the observed vapour pressure exceeds Raoult's-law prediction.

Question 8 [H] · Original PDF p. 1

Answer. Negative deviation indicates stronger A–B attractions. Mixing is generally exothermic, $\Delta H_{mix}<0$, and contraction commonly gives $\Delta V_{mix}<0$.

Question 9 [M] · Original PDF p. 1

Answer. Molarity M=moles solute/L solution; molality m=moles solute/kg solvent; mole fraction $x_i=n_i/\Sigma n$; mass percent=100(mass solute/mass solution).

Question 10 [H] · Original PDF p. 1

Answer. Molality is based on solvent mass and is essentially independent of temperature, unlike molarity, whose volume basis changes with temperature.

Question 11 [M] · Original PDF p. 1

Answer. n(NaCl)=5.85/58.44=0.1001 mol; V=0.500 L. M=0.1001/0.500=0.200 mol L^{-1} (3 s.f.).

Question 12 [H] · Original PDF p. 1

Answer. n(glucose)=18/180=0.100 mol; n(H_2O)=180/18=10.0 mol. $x_{glucose}=0.100/10.10=0.00990$; $x_{water}=10.0/10.10=0.99010$.

Question 13 [M] · Original PDF p. 1

Answer. Solubility is the amount of solute that dissolves in a specified amount of solvent at a stated temperature (and pressure for gases) to form a saturated solution. Most solids become more soluble as temperature rises, but exceptions occur.

Question 14 [M] · Original PDF p. 1

Answer. Henry's law: at fixed temperature, dissolved-gas mole fraction is proportional to its partial pressure: $p=k_H\,x$.

Question 15 [H] · Original PDF p. 1

Answer. Gas dissolution is usually exothermic. Heating favours gas escape (Le Châtelier), so gas solubility generally falls as temperature rises.

Question 16 [M] · Original PDF p. 1

Answer. High CO_2 pressure increases its dissolved mole fraction by Henry's law; opening the bottle lowers pressure and the gas escapes as bubbles.

Question 17 [H] · Original PDF p. 1

Answer. At fixed temperature, Henry's law gives $x=p/k_H$; dissolved-gas mole fraction rises linearly with partial pressure in the dilute range.

Question 18 [M] · Original PDF p. 1

Answer. A solid solution is a homogeneous solid mixture. Example: substitutional Cu–Ni alloy, where atoms of one metal replace atoms of the other in the lattice.

Question 19 [M] · Original PDF p. 2

Answer. Colligative properties depend on the number of dissolved particles, not their identity. The four are relative lowering of vapour pressure, boiling-point elevation, freezing-point depression and osmotic pressure.

Question 20 [H] · Original PDF p. 2

Answer. In dilute ideal solutions, solvent chemical potential changes according to solute-particle mole fraction. Therefore effects scale with particle count; dissociation/association changes the count and is captured by i.

Question 21 [M] · Original PDF p. 2

Answer. Relative lowering is $(p^\circ-p)/p^\circ$, where p° is pure-solvent vapour pressure and p is solution vapour pressure. For an ideal solution with non-volatile solute it equals x_{solute}.

Question 22 [M] · Original PDF p. 2

Answer. Boiling-point elevation is $\Delta T_b=T_b(\text{solution})-T_b^\circ$; for a dilute solution, $\Delta T_b=K_b\,m$ (or iK_{bm} for an electrolyte).

Question 23 [M] · Original PDF p. 2

Answer. Freezing-point depression is $\Delta T_f=T_f^\circ-T_f(\text{solution})$; for a dilute solution, $\Delta T_f=K_f\,m$ (or iK_{fm} for an electrolyte).

Question 24 [M] · Original PDF p. 2

Answer. Osmotic pressure π is the minimum external pressure applied to a solution to stop net solvent flow through a semipermeable membrane.

Question 25 [H] · Original PDF p. 2

Answer. At equilibrium, $\mu_A^*(P)=\mu_A(\text{solution},P+\pi)\approx\mu_A^*(P)+V_A\pi+RT\ln x_A$, so $\pi=-(RT/V_A)\ln x_A$. For a dilute solution, $-\ln x_A\approx x_B\approx n_B/n_A$ and $V\approx n_{AVA}$; hence $\pi V\approx n_{BRT}$, or $\pi=CRT$ with $C=n_B/V$. Assumptions: ideal dilute solution, small pressure correction, constant solvent partial molar volume, solvent-only membrane, uniform T, and negligible solute volatility.

Question 26 [M] · Original PDF p. 2

Answer. For dilute polymer solutions, π is measurable at low concentration without heating or freezing the macromolecule. From $\pi V=nRT$, its molar mass can be obtained even when other colligative changes are very small.

Question 27 [H] · Original PDF p. 2

Answer. Osmosis is net solvent flow from lower solute concentration (lower π) to higher π through a semipermeable membrane. Reverse osmosis applies pressure greater than π to reverse that flow; it is used for desalination.

Question 28 [M] · Original PDF p. 2

Answer. Isotonic solutions have equal osmotic pressure at the same T. Hypertonic has higher π than the reference; hypotonic has lower π. Solvent tends to flow toward the hypertonic side.

Question 29 [H] · Original PDF p. 2

Answer. Solvent flows from the solution of lower osmotic pressure to that of higher osmotic pressure until equilibrium, provided the membrane is selectively permeable to solvent.

Question 30 [M] · Original PDF p. 2

Answer. An abnormal molar mass is an experimentally inferred value that differs from the formula molar mass because solute particles associate or dissociate in solution.

Question 31 [M] · Original PDF p. 2

Answer. The van't Hoff factor i is observed colligative effect divided by the value calculated for a non-associating, non-dissociating solute at the same concentration. Equivalently, $i = M_{formula}/M_{observed}$.

Question 32 [H] · Original PDF p. 2

Answer. Association reduces the number of solute particles, so $i<1$. Since $M_{observed} = M_{formula}/i$, the apparent molar mass is greater than the formula molar mass.

Question 33 [H] · Original PDF p. 2

Answer. Dissociation increases particle count, so $i>1$. Consequently $M_{observed} = M_{formula}/i$ is smaller than the formula molar mass.

Question 34 [M] · Original PDF p. 2

Answer. In dilute water NaCl dissociates mainly into Na^+ and Cl^-, increasing the number of particles; hence i is greater than one (approaching 2 at ideal infinite dilution).

Question 35 [H] · Original PDF p. 2

Answer. For initial N molecules, dimerisation of fraction α leaves $N(1-\alpha)+N\alpha/2$ particles. Thus $i = 1-\alpha/2$.

Question 36 [H] · Original PDF p. 2

Answer. If one solute unit produces n particles upon dissociation, initial N units yield $N(1-\alpha)+n N\alpha$ particles. Therefore $i = 1+\alpha(n-1)$.

Question 37 [M] · Original PDF p. 2

Answer. Required data: solute mass, solvent mass, K_f, measured ΔT_f, and confirmation that $i=1$ (or an independently known i). Then $m = \Delta T_f/(iK_f)$ and molar mass = solute mass/(molality × kg solvent).

Question 38 [H] · Original PDF p. 2

Answer. For a non-electrolyte, $\Delta T_f = K_{fm} = (1.86\ \text{K kg mol}^{-1})(0.20\ \text{mol kg}^{-1}) = 0.372\ \text{K}$.

Question 39 [H] · Original PDF p. 3

Answer. $i = \Delta T_f/(K_{fm}) = 0.279/(1.86\times0.10) = 1.50$.

Question 40 [H] · Original PDF p. 3

Answer. $n = \pi V/RT = (0.0246\ \text{atm}\times0.100\ \text{L})/(0.0821\ \text{L atm mol}^{-1}\ \text{K}^{-1}\times300\ \text{K}) = 9.99\times10^{-5}$ mol. $M = 1.0\ \text{g}/n \approx 1.00\times10^{4}\ \text{g mol}^{-1}$.

Question 41 [M] · Original PDF p. 3

Answer. A non-volatile solute lowers solvent vapour pressure. The solution must be heated to a higher temperature to reach the external pressure at boiling.

Question 42 [H] · Original PDF p. 3

Answer. Ethylene glycol lowers the mole fraction and chemical potential of water; the liquid must be cooled further to reach equilibrium with ice, so its freezing point falls.

Question 43 [M] · Original PDF p. 3

Answer. For equal molality in the same solvent, $\Delta T_b/\Delta T_f = K_b/K_f$. The two temperature changes are not generally equal because the solvent constants differ.

Question 44 [H] · Original PDF p. 3

Answer. Positive deviation implies weaker solute–solvent attractions than the average like-component attractions; molecules escape more readily and vapour pressure is higher.

Question 45 [M] · Original PDF p. 3

Answer. Ideal solutions obey Raoult's law throughout and have $\Delta H_{mix} = \Delta V_{mix} = 0$. Non-ideal solutions deviate because unlike interactions differ; ΔH_{mix} and/or ΔV_{mix} are non-zero.

Question 46 [H] · Original PDF p. 3

Answer. Mixing acetone disrupts some ethanol–ethanol hydrogen bonding, and acetone–ethanol attractions are comparatively weaker than the original interactions; this can produce positive deviation.

Question 47 [M] · Original PDF p. 3

Answer. An azeotrope boils at fixed composition because liquid and vapour compositions match, so it distils without composition change. Positive deviation gives a minimum-boiling azeotrope; negative deviation gives a maximum-boiling one.

Question 48 [H] · Original PDF p. 3

Answer. At the azeotropic composition, vapour has the same composition as liquid. Fractional distillation cannot enrich one component past that composition at the azeotropic pressure.

Question 49 [M] · Original PDF p. 3

Answer. Increasing temperature raises the fraction of molecules with enough energy to escape the liquid, so equilibrium vapour pressure increases.

Question 50 [H] · Original PDF p. 3

Answer. Mole fraction is based on moles, not mass. For equal masses, $n = m/M$; the liquid with lower molar mass contributes more moles, so the fractions generally differ.

Question 51 [M] · Original PDF p. 3

Answer. Mass percent $= 100 \times 12/(12+108) = 10.0\%$ (w/w).

Question 52 [H] · Original PDF p. 3

Answer. moles solute $= (2.0 \text{ mol kg}^{-1})(0.500 \text{ kg water}) = 1.00$ mol.

Question 53 [M] · Original PDF p. 3

Answer. 10% (w/w) means 10 g solute in every 100 g of total solution (90 g solvent for a binary solution).

Question 54 [H] · Original PDF p. 3

Answer. $n(\text{urea}) = 5/60.06 = 0.0833$ mol; solvent $= 0.095$ kg. Molality $= 0.0833/0.095 = 0.877 \text{ mol kg}^{-1}$.

Question 55 [M] · Original PDF p. 3

Answer. For each volatile component, $p_i = x_i\, p_i^\circ$ in an ideal solution. Changing its mole fraction changes its partial pressure and therefore the total vapour pressure.

Question 56 [H] · Original PDF p. 4

Answer. For ideal binary liquids A and B: $p_A = x_{ApA}{}^\circ$, $p_B = x_{BpB}{}^\circ$; $P_{total} = x_{ApA}{}^\circ + x_{BpB}{}^\circ$.

Question 57 [M] · Original PDF p. 4

Answer. For P_{total} versus x_A, slope is $p_A{}^\circ - p_B{}^\circ$. Partial-pressure lines have slopes $+p_A{}^\circ$ for A and $-p_B{}^\circ$ for B when plotted against x_A.

Question 58 [H] · Original PDF p. 4

Answer. If unlike A–B attractions are weaker, molecules escape more readily (positive deviation); if stronger, escape is suppressed (negative deviation). Heat/volume changes often have matching signs but are not the defining criterion.

Question 59 [M] · Original PDF p. 4

Answer. Higher pressure increases gas chemical potential above the liquid and drives more gas into solution; at fixed T the proportionality is Henry's law.

Question 60 [H] · Original PDF p. 4

Answer. $x = p/k_H = 50\ \text{kPa}/(1.0\times10^{5}\ \text{kPa}) = 5.0\times10^{-4}$.

Question 61 [M] · Original PDF p. 4

Answer. Molality is moles solute per kilogram solvent and depends on solvent mass. Mole fraction is moles of one component divided by total moles and is dimensionless.

Question 62 [H] · Original PDF p. 4

Answer. $\Delta T_b = K_{bm} = (0.512\ \text{K kg mol}^{-1})(0.50\ \text{mol kg}^{-1}) = 0.256\ \text{K}$.

Question 63 [M] · Original PDF p. 4

Answer. A semipermeable membrane must pass solvent but retain solute; otherwise no selective solvent flow or measurable osmotic pressure develops.

Question 64 [H] · Original PDF p. 4

Answer. $i = \pi/(CRT) = 1.23/[0.050\times0.0821\times300] = 0.999 \approx 1.00$.

Question 65 [M] · Original PDF p. 4

Answer. It requires precise temperature measurement and enough sample; small ΔT_b may be difficult to resolve. Volatile or thermally unstable solutes/solvents can also invalidate the method.

Question 66 [H] · Original PDF p. 4

Answer. For dissociation into 3 particles, $i = 1+\alpha(n-1) = 1+0.60(2) = 2.20$.

Question 67 [M] · Original PDF p. 4

Answer. $i=1$ indicates the effective particle count equals the formula-unit count: negligible association/dissociation in the ideal dilute limit.

Question 68 [H] · Original PDF p. 4

Answer. For trimerisation, $i = 1-2\alpha/3$ because each associated group of three replaces three particles by one.

Question 69 [M] · Original PDF p. 4

Answer. The method requires heating to boiling; a thermally unstable solute may decompose or react, so the measured boiling point would not represent the original solution.

Question 70 [H] · Original PDF p. 4

Answer. Association gives $i<1$ and $M_{observed} > M_{formula}$; dissociation gives $i>1$ and $M_{observed} < M_{formula}$. The direction and size of the discrepancy diagnose particle-number change.

Question 71 [M] · Original PDF p. 4

Answer. Each volatile component contributes a partial pressure; for an ideal mixture $p_i = x_i\, p_i^{\circ}$, and Dalton's law gives $P_{total} = \Sigma_i\, p_i$.

Question 72 [H] · Original PDF p. 4

Answer. $CaCl_2 \rightarrow Ca^{2+} + 2Cl^-$, so $i \approx 3$. $\Delta T_b = iK_{bm} = 3\times0.512\times0.20 = 0.307$ K (complete dissociation, dilute-solution assumption).

UNIT 2: ELECTROCHEMISTRY

Questions 73–144 · Original PDF pages 5–8

Question 73 [M] · Original PDF p. 5

Answer. Oxidation is loss of electrons; reduction is gain of electrons. They occur together in a redox reaction.

Question 74 [M] · Original PDF p. 5

Answer. A galvanic cell uses a spontaneous redox reaction to produce electrical energy (anode negative); an electrolytic cell uses an external supply to drive a non-spontaneous reaction (anode positive). Oxidation is at the anode and reduction at the cathode in both.

Question 75 [M] · Original PDF p. 5

Answer. EMF is the open-circuit potential difference between the cell terminals, equal to the maximum electrical work per unit charge under specified conditions.

Question 76 [M] · Original PDF p. 5

Answer. Standard electrode potential is a reduction potential measured relative to the standard hydrogen electrode under standard conditions: solute activities 1 (approximately 1 M), gases 1 bar, pure solids/liquids, specified temperature (usually 298 K).

Question 77 [H] · Original PDF p. 5

Answer. An absolute single-electrode potential cannot be measured independently; only cell differences are measurable. Assigning the SHE zero is a reference convention that allows all other electrode potentials to be tabulated consistently.

Question 78 [M] · Original PDF p. 5

Answer. Daniell cell: $Zn(s)|Zn^{2+}(aq, 1\ M)||Cu^{2+}(aq, 1\ M)|Cu(s)$, with the anode on the left.

Question 79 [H] · Original PDF p. 5

Answer. Maximum non-expansion electrical work is $-nFE_{cell}$. At reversible standard conditions $\Delta G^\circ = w_{max}$, so $\Delta G^\circ = -nFE^\circ_{cell}$.

Question 80 [M] · Original PDF p. 5

Answer. For a balanced cell reaction, $E = E^\circ - (RT/nF)\ln Q$. At 298 K, $E = E^\circ - (0.05916\ V/n)\log_{10} Q$.

Question 81 [H] · Original PDF p. 5

Answer. $Zn(s) + Cu^{2+} \rightarrow Zn^{2+} + Cu(s)$, $Q = a(Zn^{2+})/a(Cu^{2+})$. $E = E^\circ - (RT/2F)\ln([Zn^{2+}]/[Cu^{2+}])$; at 298 K use 0.05916/2 times $\log_{10}$ of that ratio.

Question 82 [M] · Original PDF p. 5

Answer. $\Delta G = -nFE_{cell}$: $E > 0$ means $\Delta G < 0$ and forward reaction spontaneous; $E = 0$ is equilibrium; $E < 0$ means the forward reaction is non-spontaneous as written.

Question 83 [H] · Original PDF p. 5

Answer. $\Delta G^\circ = -nFE^\circ = -(2)(96500)(1.10) = -2.123 \times 10^5$ J mol^{-1} $= -212.3$ kJ mol^{-1}.

Question 84 [M] · Original PDF p. 5

Answer. For the balanced cell reaction, K is the equilibrium ratio of product activities to reactant activities, each raised to its stoichiometric coefficient; pure solids and liquids are omitted.

Question 85 [H] · Original PDF p. 5

Answer. At equilibrium $E = 0$ and $Q = K$. The Nernst equation gives $E^\circ = (RT/nF)\ln K$; at 298 K, $E^\circ = (0.05916\ V/n)\log_{10} K$.

Question 86 [M] · Original PDF p. 5

Answer. The salt bridge completes the ionic circuit and maintains electroneutrality by ion migration, while limiting direct mixing of the half-cell solutions.

Question 87 [H] · Original PDF p. 5

Answer. At equilibrium Q=K, so E=0 and ΔG=0. There is no net thermodynamic driving force for electron flow or net cell reaction.

Question 88 [M] · Original PDF p. 5

Answer. Specific conductivity κ is conductance of a solution column of unit length and unit cross-sectional area (κ=G l/A). SI unit: S m^{-1}.

Question 89 [M] · Original PDF p. 5

Answer. Molar conductivity Λ_m is conductivity multiplied by volume containing one mole of electrolyte. Usual unit: S cm^2 mol^{-1} (SI: S m^2 mol^{-1}).

Question 90 [H] · Original PDF p. 5

Answer. Dilution reduces the number of ions per unit volume, so κ falls. Per mole, ions experience less interionic restraint and weak electrolytes ionise more, so Λ_m rises.

Question 91 [M] · Original PDF p. 5

Answer. For strong electrolytes Λ_m rises modestly on dilution as interionic effects diminish; for weak electrolytes it rises sharply because dilution also increases ionisation.

Question 92 [H] · Original PDF p. 5

Answer. A weak electrolyte is only partly ionised when concentrated; dilution shifts its ionisation equilibrium toward ions, increasing the number of charge carriers and Λ_m substantially.

Question 93 [M] · Original PDF p. 5

Answer. At infinite dilution each ion contributes independently to total molar conductivity: $\Lambda_m^\circ = \nu_+\lambda_+^\circ + \nu_-\lambda_-^\circ$, with stoichiometric coefficients ν_i.

Question 94 [H] · Original PDF p. 6

Answer. By ionic additivity, $\Lambda_m^\circ(CH_3COOH) = \Lambda_m^\circ(HCl) + \Lambda_m^\circ(CH_3COONa) - \Lambda_m^\circ(NaCl)$; the strong-electrolyte terms cancel Na^+ and Cl^-, leaving H^+ and acetate contributions.

Question 95 [M] · Original PDF p. 6

Answer. For a weak electrolyte, degree of dissociation $\alpha = \Lambda_m/\Lambda_m^\circ$, where Λ_m is measured at the concentration and Λ_m° is the limiting value.

Question 96 [H] · Original PDF p. 6

Answer. Measure Λ_m and obtain Λ_m° (e.g. by Kohlrausch's law); set $\alpha = \Lambda_m/\Lambda_m^\circ$. For a weak 1:1 electrolyte of concentration c, $K_a = c\alpha^2/(1-\alpha)$.

Question 97 [M] · Original PDF p. 6

Answer. Faraday's first law: the mass deposited or liberated at an electrode is directly proportional to the charge passed, $m \propto Q$.

Question 98 [M] · Original PDF p. 6

Answer. Faraday's second law: for the same charge through different electrolytes, deposited masses are proportional to their chemical equivalent masses (molar mass divided by electron number).

Question 99 [H] · Original PDF p. 6

Answer. $Ag^+ + e^- \rightarrow Ag$ requires one mole of electrons per mole Ag. $Q = F = 9.65\times10^4$ C to deposit one mole.

Question 100 [H] · Original PDF p. 6

Answer. $Q = It = 2.0\times(30\times60) = 3600$ C. $n(Cu) = Q/(2F) = 0.01865$ mol; $m = 0.01865\times63.5 = 1.18$ g.

Question 101 [M] · Original PDF p. 6

Answer. The electrolyte supplies mobile ions to carry current, completes ionic conduction, and undergoes the electrode reactions; it must be sufficiently ionically conducting.

Question 102 [H] · Original PDF p. 6

Answer. Molten NaCl: Na^+ is reduced to Na(l) at the cathode; Cl^- is oxidised to $Cl_2(g)$ at the anode. Aqueous NaCl: water is reduced to H_2 at the cathode and NaOH remains; concentrated brine gives Cl_2 at the anode, while very dilute chloride can give O_2 instead.

Question 103 [M] · Original PDF p. 6

Answer. A Leclanché dry cell has a zinc container (anode), central carbon rod with MnO_2/carbon depolariser (cathode mixture), and moist $NH_4Cl/ZnCl_2$ paste electrolyte. Zn oxidation and cathodic reduction generate current; it is a primary, non-rechargeable cell.

Question 104 [M] · Original PDF p. 6

Answer. A lead accumulator has a Pb negative plate, PbO_2 positive plate, porous separators and aqueous H_2SO_4 electrolyte in a container. Both plates become $PbSO_4$ during discharge; charging reverses the reaction.

Question 105 [H] · Original PDF p. 6

Answer. Negative plate: $Pb+SO_4^{2-}\rightarrow PbSO_4+2e^-$. Positive plate: $PbO_2+4H^++SO_4^{2-}+2e^-\rightarrow PbSO_4+2H_2O$. Overall: $Pb+PbO_2+2H_2SO_4\rightarrow 2PbSO_4+2H_2O$.

Question 106 [M] · Original PDF p. 6

Answer. A fuel cell continuously converts supplied fuel and oxidant directly to electrical energy through redox reactions. An advantage is high efficiency with low local pollutant emission (for H_2/O_2, water is the product).

Question 107 [H] · Original PDF p. 6

Answer. In alkaline electrolyte: anode $H_2+2OH^-\rightarrow 2H_2O+2e^-$; cathode $\frac{1}{2}O_2+H_2O+2e^-\rightarrow 2OH^-$. Overall $H_2+\frac{1}{2}O_2\rightarrow H_2O$.

Question 108 [M] · Original PDF p. 6

Answer. Corrosion involves spatially separated anodic metal oxidation and cathodic reduction reactions on a conducting wet surface, coupled by electron and ion transport—an electrochemical cell.

Question 109 [H] · Original PDF p. 6

Answer. At anodic sites $Fe\rightarrow Fe^{2+}+2e^-$. At aerated cathodic sites $O_2+2H_2O+4e^-\rightarrow 4OH^-$. Fe^{2+} forms $Fe(OH)_2$, then oxidises/hydrates to Fe(III) oxyhydroxides and hydrated rust, often represented as $Fe_2O_3\cdot xH_2O$.

Question 110 [M] · Original PDF p. 6

Answer. Primary batteries are normally non-rechargeable because their cell reaction is not practically reversible. Secondary batteries can be recharged by driving the reverse reaction with an external current.

Question 111 [H] · Original PDF p. 6

Answer. $\Delta G^\circ=-nFE^\circ$. Since $E^\circ=+0.46$ V and $n>0$, $\Delta G^\circ<0$; the reaction as written is standard-state spontaneous.

Question 112 [M] · Original PDF p. 6

Answer. The electrochemical series ranks redox couples by standard reduction potential. It helps predict cell EMFs/spontaneity and compare oxidising or reducing strengths, including likely displacement reactions.

Question 113 [H] · Original PDF p. 6

Answer. Compare standard reduction potentials: the species with the more positive E° is reduced at the cathode. A metal M can displace N^{2+} if $E^\circ_{cell}=E^\circ(N^{2+}/N)-E^\circ(M^{2+}/M)>0$ for $M+N^{2+}\rightarrow M^{2+}+N$.

Question 114 [M] · Original PDF p. 6

Answer. It is measured relative to the standard hydrogen electrode (SHE), whose potential is defined as zero under standard conditions.

Question 115 [H] · Original PDF p. 6

Answer. As reaction proceeds, ion activities change and hence Q changes. The Nernst equation $E=E^\circ-(RT/nF)\ln Q$ therefore makes the cell potential vary until it reaches zero at equilibrium.

Question 116 [M] · Original PDF p. 6

Answer. Limiting molar conductivity Λ_m° is the molar conductivity extrapolated to infinite dilution, where interionic interactions vanish and ions migrate independently.

Question 117 [H] · Original PDF p. 6

Answer. Ionic mobility is drift velocity per unit electric field for a particular ion. Conductivity is the bulk ability of all ions in solution to carry current and depends on concentration, charge and mobility.

Question 118 [M] · Original PDF p. 7

Answer. Higher temperature generally lowers solvent viscosity and increases ionic mobility; it may also increase weak-electrolyte ionisation, raising conductivity.

Question 119 [H] · Original PDF p. 7

Answer. κ=cell constant/R=1.25 cm^{-1}/200 Ω=0.00625 S cm^{-1}.

Question 120 [H] · Original PDF p. 7

Answer. $\Lambda_m=\kappa\times1000/c=(1.5\times10^{-3}\ S\ cm^{-1}\times1000\ cm^3\ L^{-1})/(0.01\ mol\ L^{-1})=150\ S\ cm^2\ mol^{-1}$.

Question 121 [M] · Original PDF p. 7

Answer. Metallic conduction is electron motion without bulk chemical change. Electrolytic conduction is ion motion in solution or melt and is accompanied by electrode redox reactions.

Question 122 [H] · Original PDF p. 7

Answer. For $n>0$, $\Delta G^\circ=-nFE^\circ=0$. Also $\Delta G^\circ=-RT\ \ln K$, hence $K=1$ at that temperature under standard-state conventions.

Question 123 [M] · Original PDF p. 7

Answer. Electrons flow through the external circuit from the zinc anode to the copper cathode.

Question 124 [H] · Original PDF p. 7

Answer. $E^\circ cell=E^\circ cathode-E^\circ anode=0.34-(-0.76)=1.10$ V.

Question 125 [M] · Original PDF p. 7

Answer. Anode is where oxidation supplies electrons in a galvanic cell, making it negative. In an electrolytic cell the power supply removes electrons from the anode, making it positive; oxidation still occurs there.

Question 126 [H] · Original PDF p. 7

Answer. For a two-electron cell reaction, $\Delta G^\circ=-2FE^\circ cell$ (per mole of reaction as written).

Question 127 [M] · Original PDF p. 7

Answer. Anions from the bridge migrate toward the anode compartment, where positive charge is produced; cations migrate toward the cathode compartment, where cations are consumed, maintaining bulk neutrality.

Question 128 [H] · Original PDF p. 7

Answer. For $Ox+ne^-\rightarrow Red$, $Q=a(Red)/a(Ox)$; increasing Ox lowers Q and raises E by Nernst. If the oxidised species occurs on the product side of the chosen cell reaction, the direction reverses; the balanced reaction must be specified.

Question 129 [M] · Original PDF p. 7

Answer. Q is the ratio of instantaneous product activities to reactant activities, each raised to its stoichiometric coefficient; pure solids and liquids are omitted. Unlike K, it need not describe equilibrium.

Question 130 [H] · Original PDF p. 7

Answer. $E=E^\circ-(0.05916/2)\log_{10}(100)=0.50-0.05916=0.44084$ V$\approx$0.441 V.

Question 131 [M] · Original PDF p. 7

Answer. Molar conductivity normalises conductivity to the amount of electrolyte, allowing comparison of ion-transport contribution per mole at different concentrations.

Question 132 [H] · Original PDF p. 7

Answer. At finite concentration, oppositely charged ions attract and like-charged ions repel, hindering independent motion and lowering ionic mobility and Λ_m; dilution reduces these effects.

Question 133 [M] · Original PDF p. 7

Answer. Cell constant is electrode separation divided by electrode area, l/A (cm^{-1}). Determine it by measuring resistance/conductance of a standard solution with known κ.

Question 134 [H] · Original PDF p. 7

Answer. $\kappa = G \times \text{cell constant} = (0.005\ S)(0.8\ cm^{-1}) = 0.0040\ S\ cm^{-1}$.

Question 135 [M] · Original PDF p. 7

Answer. For a weak electrolyte, ionisation is an equilibrium; dilution favours the side with more dissolved particles, increasing the fraction ionised (Ostwald dilution law).

Question 136 [H] · Original PDF p. 8

Answer. $\Lambda^\circ_m(CH_3COOH) = \Lambda^\circ_m(HCl) + \Lambda^\circ_m(CH_3COONa) - \Lambda^\circ_m(NaCl)$.

Question 137 [M] · Original PDF p. 8

Answer. Electrochemical equivalent Z is mass deposited/liberated per unit charge passed: $Z = m/Q$; SI unit kg C^{-1}.

Question 138 [H] · Original PDF p. 8

Answer. $n(Ag) = 1.08/108 = 0.0100$ mol, so $Q = 0.0100F = 965$ C. At 1.0 A, $t = Q/I = 965$ s (about 16.1 min).

Question 139 [M] · Original PDF p. 8

Answer. Overvoltage (overpotential) is the extra potential beyond the reversible thermodynamic electrode potential needed to sustain electrolysis at a stated current density, due to kinetic and transport effects.

Question 140 [H] · Original PDF p. 8

Answer. In aqueous Na_2SO_4, water is reduced at the cathode to H_2 and oxidised at the anode to O_2; Na^+ and SO_4^{2-} mainly maintain conductivity. Water reduction is favoured over $Na^+ \rightarrow Na$ because sodium metal would immediately react with water.

Question 141 [M] · Original PDF p. 8

Answer. PbO_2 is the positive plate's oxidising active material: it accepts electrons during discharge and is reduced to $PbSO_4$, giving the cell a useful positive electrode potential.

Question 142 [H] · Original PDF p. 8

Answer. A hydrogen fuel cell spontaneously converts chemical energy of continuously supplied H_2/O_2 to electricity and water. An electrolytic cell consumes electrical energy to drive a non-spontaneous chemical change.

Question 143 [M] · Original PDF p. 8

Answer. Potential is energy per unit charge (J C^{-1}), independent of the total amount of electrode material or reaction; it is an intensive quantity.

Question 144 [H] · Original PDF p. 8

Answer. Different ion activities create a chemical-potential gradient. The dilute side is the anode (metal oxidises, raising its ion concentration); the concentrated side is the cathode (ions reduce and plate). Electrons flow dilute→concentrated externally. For M^{z+}, $E = (RT/zF)\ln(a_{high}/a_{low})$; no unique numeric value is possible without z and activity data.

UNIT 3: CHEMICAL KINETICS

Questions 145–216 · Original PDF pages 9–12

Question 145 [M] · Original PDF p. 9

Answer. Reaction rate is change in concentration of a reactant or product per unit time, with stoichiometric normalization and a negative sign for reactants so the rate is positive.

Question 146 [M] · Original PDF p. 9

Answer. Average rate is concentration change divided by a finite time interval. Instantaneous rate is the derivative d(concentration)/dt at a particular time (with stoichiometric sign/normalization).

Question 147 [H] · Original PDF p. 9

Answer. For aA→bB, rate=−(1/a)d[A]/dt=(1/b)d[B]/dt; stoichiometric coefficients make the rates calculated from each species equal.

Question 148 [M] · Original PDF p. 9

Answer. Rate law is an experimentally determined relation such as rate=k[A]^m[B]^n. The rate constant k is the proportionality coefficient at a specified temperature; its units depend on overall order.

Question 149 [M] · Original PDF p. 9

Answer. Reaction order is the sum of the experimentally measured exponents in the rate law; individual orders are the exponents of the corresponding concentrations.

Question 150 [M] · Original PDF p. 9

Answer. Molecularity is the number of reacting entities in a single elementary step and is a positive integer. Order is the sum of empirical rate-law exponents and applies to an overall reaction or step; it may be zero or fractional.

Question 151 [H] · Original PDF p. 9

Answer. Order is inferred from measured rate dependence and can be any real value, including zero or negative. Molecularity counts discrete species colliding in one elementary event, so it is a positive integer.

Question 152 [M] · Original PDF p. 9

Answer. A pseudo-first-order reaction is actually of higher order but behaves first order when one reactant is in large excess and effectively constant; e.g. ester hydrolysis in excess water.

Question 153 [H] · Original PDF p. 9

Answer. Water is present in huge excess, so its concentration changes negligibly. In rate=k[ester][H_2O], absorb the nearly constant [H_2O] into k_{obs}=k[H_2O], giving rate=k_{obs}[ester].

Question 154 [M] · Original PDF p. 9

Answer. For A→products at zero order: $[A]_t=[A]_0-kt$ (valid until reactant is depleted).

Question 155 [M] · Original PDF p. 9

Answer. For A→products at first order: $\ln([A]_0/[A]_t)=kt$, or $[A]_t=[A]_{0e}$^(−kt).

Question 156 [M] · Original PDF p. 9

Answer. For a zero-order reaction, $t_1/2=[A]_0/(2k)$.

Question 157 [M] · Original PDF p. 9

Answer. For a first-order reaction, $t_1/2=\ln 2/k=0.693/k$.

Question 158 [H] · Original PDF p. 9

Answer. At half-life, $[A]_t=[A]_0/2$. Substitute into $\ln([A]_0/[A]_t)=kt$: $\ln 2=kt_1/2$, so $t_1/2=0.693/k$, independent of $[A]_0$.

Question 159 [H] · Original PDF p. 9

Answer. 75% complete means 25% remains, equal to one quarter of the initial amount: two half-lives have elapsed. Thus $t_1/2=40/2=20$ min.

Question 160 [H] · Original PDF p. 9

Answer. $k=(1/t)\ln([A]_0/[A]_t)=\ln(1/0.20)/30=\ln 5/30=0.0536\ \text{min}^{-1}$.

Question 161 [H] · Original PDF p. 9

Answer. For zero order, $[A]_t=[A]_0-kt$. $t=(0.10-0.04)/(2.0\times10^{-3})=30$ s.

Question 162 [M] · Original PDF p. 9

Answer. Zero-order k has concentration/time units, e.g. mol L^{-1} s^{-1} (M s^{-1}).

Question 163 [M] · Original PDF p. 9

Answer. First-order k has units of inverse time, e.g. s^{-1} or min^{-1}.

Question 164 [H] · Original PDF p. 9

Answer. A straight-line plot of [A] versus t identifies zero-order behaviour; its slope is $-k$ and its intercept is $[A]_0$.

Question 165 [H] · Original PDF p. 9

Answer. A straight-line plot of ln[A] versus t identifies first-order behaviour; its slope is $-k$ and its intercept is $\ln[A]_0$.

Question 166 [M] · Original PDF p. 10

Answer. Arrhenius equation: $k=Ae^{(-E_a/RT)}$, or $\ln k=\ln A-E_a/(RT)$, where A is the frequency factor.

Question 167 [M] · Original PDF p. 10

Answer. Activation energy E_a is the energy barrier between reactants and the activated state along a reaction pathway; it is the minimum additional energy required in the simple Arrhenius picture.

Question 168 [H] · Original PDF p. 10

Answer. A catalyst provides a pathway with lower activation barriers for both forward and reverse reactions. It speeds both directions but leaves ΔG° and K unchanged, so equilibrium composition is unchanged.

Question 169 [M] · Original PDF p. 10

Answer. Collision theory models reaction as molecular collisions; only collisions with sufficient energy and suitable orientation are effective and form products.

Question 170 [H] · Original PDF p. 10

Answer. Most collisions lack energy at least equal to the threshold/activation energy or have unsuitable orientation for bond rearrangement, so they do not form products.

Question 171 [M] · Original PDF p. 10

Answer. An effective collision has sufficient energy to overcome the activation barrier and an orientation that permits the required bonds to break/form.

Question 172 [H] · Original PDF p. 10

Answer. Even energetic molecules may not react if reactive sites do not meet in the proper geometry. Correct orientation is represented by a steric/orientation factor in collision theory.

Question 173 [M] · Original PDF p. 10

Answer. Increasing reactant concentration generally increases collision frequency and, for positive-order species, raises the rate; the exact dependence is set by the rate law.

Question 174 [M] · Original PDF p. 10

Answer. Increasing temperature generally raises the rate constant because a larger fraction of molecules has energy at or above E_a; rates usually increase substantially.

Question 175 [H] · Original PDF p. 10

Answer. The Arrhenius factor e^(−E_a/RT) changes exponentially with 1/T; heating appreciably increases the fraction of molecules able to cross the barrier, especially for large E_a.

Question 176 [H] · Original PDF p. 10

Answer. Use $\ln(k_2/k_1)=E_a/R(1/T_1-1/T_2)$. $E_a=8.314\ \ln 2/(1/300-1/310)=5.36\times10^4$ J mol$^{-1}\approx$53.6 kJ mol^{-1}.

Question 177 [M] · Original PDF p. 10

Answer. A is the Arrhenius frequency factor (pre-exponential factor), reflecting collision frequency and orientation probability; it has the same units as k.

Question 178 [H] · Original PDF p. 10

Answer. From $\ln k=\ln A-E_a/(RT)$, subtract values at T_1 and T_2: $\ln(k_2/k_1)=-E_a/R(1/T_2-1/T_1)=E_a/R(1/T_1-1/T_2)$.

Question 179 [M] · Original PDF p. 10

Answer. For ln k versus 1/T, slope=$-E_a/R$ and intercept=ln A. The plot gives activation energy and frequency factor.

Question 180 [H] · Original PDF p. 10

Answer. Find slope m of ln k versus 1/T; $E_a=-mR$. If $\log_{10}k$ is plotted instead, $E_a=-2.303Rm$.

Question 181 [M] · Original PDF p. 10

Answer. A slow rate is an observed outcome that can also reflect low concentration or an unfavourable mechanism/pre-exponential factor. High E_a is a property of a pathway's energy barrier and often causes low rates at a given T, but the terms are not identical.

Question 182 [H] · Original PDF p. 10

Answer. An overall stoichiometric equation does not reveal the elementary steps or intermediates. Therefore rate exponents cannot generally be inferred from its coefficients and must be measured experimentally (except for an elementary step).

Question 183 [M] · Original PDF p. 10

Answer. Overall order=2+1=3 (second order in A and first order in B).

Question 184 [H] · Original PDF p. 10

Answer. Rate factor=$(2^2)(3^1)=12$; doubling [A] and tripling [B] makes the rate twelve times larger.

Question 185 [M] · Original PDF p. 10

Answer. The rate-determining step is the slow step that constrains the overall reaction rate under the stated conditions; its position and preceding equilibria determine the observed law.

Question 186 [H] · Original PDF p. 10

Answer. An elementary step is a single molecular event, so its molecularity specifies the colliding species and its rate law. An overall equation combines multiple steps and its coefficients need not be the kinetic exponents.

Question 187 [M] · Original PDF p. 10

Answer. The overall reaction is the net stoichiometric change. A mechanism is the sequence of elementary steps, intermediates and transition states that produces that net change.

Question 188 [H] · Original PDF p. 10

Answer. $t_1/2=0.693/k=0.693/(6.93\times10^{-3}\ \text{min}^{-1})=100$ min.

Question 189 [M] · Original PDF p. 10

Answer. After two half-lives, fraction remaining=$(1/2)^2=1/4=25\%$.

Question 190 [H] · Original PDF p. 11

Answer. For zero order, $k=([A]_0-[A]_t)/t=(0.10-0.06)/20=0.0020$ M min^{-1}.

Question 191 [M] · Original PDF p. 11

Answer. A catalyst lowers the activation energy for both forward and reverse reactions through an alternative pathway; the change in barrier is consistent with the unchanged reaction ΔG.

Question 192 [H] · Original PDF p. 11

Answer. A catalyst accelerates forward and reverse reactions and helps equilibrium be reached sooner; it does not change ΔG° or K, so it cannot shift equilibrium composition.

Question 193 [M] · Original PDF p. 11

Answer. Thermal reactions receive energy through heat, usually a thermal molecular energy distribution. Photochemical reactions are initiated by absorption of photons, often producing electronically excited species.

Question 194 [H] · Original PDF p. 11

Answer. Elementary collision theory treats molecules as simple colliding particles and is approximate for complex polyatomic molecules, solvent effects, energy redistribution, multi-step pathways and detailed orientation/entropy effects.

Question 195 [M] · Original PDF p. 11

Answer. Order in A=1; order in B=2; overall order=1+2=3.

Question 196 [H] · Original PDF p. 11

Answer. Rate factor=$(1/2)^2\times 2=1/2$. The rate becomes half its original value.

Question 197 [M] · Original PDF p. 11

Answer. At a fixed temperature and for unchanged mechanism, k is the proportionality constant set by the pathway and temperature; concentration dependence is represented separately by concentration powers in the rate law.

Question 198 [H] · Original PDF p. 11

Answer. 90% complete leaves 10%. $k=\ln(10)/50=0.0461\ \text{min}^{-1}$.

Question 199 [M] · Original PDF p. 11

Answer. After three half-lives, fraction remaining=$(1/2)^3=1/8=12.5\%$.

Question 200 [H] · Original PDF p. 11

Answer. For zero order, $t_1/2=[A]_0/(2k)=0.80/(2\times 0.02)=20$ min.

Question 201 [M] · Original PDF p. 11

Answer. In a plot of ln[A] versus time, slope=$-k$. The negative sign reflects the decrease in reactant concentration; its magnitude is the first-order rate constant.

Question 202 [H] · Original PDF p. 11

Answer. For zero order, $t_1/2=[A]_0/(2k)$, so it increases with initial concentration. For first order, $t_1/2=0.693/k$ and is independent of initial concentration.

Question 203 [M] · Original PDF p. 11

Answer. A catalyst creates an alternative pathway with a lower activation barrier (or different sequence of steps), increasing rate without changing the net reaction equilibrium.

Question 204 [H] · Original PDF p. 11

Answer. For ln k versus 1/T, slope=$-E_a/R$. For $\log_{10}k$ versus 1/T, slope=$-E_a/(2.303R)$, smaller in magnitude by factor 2.303.

Question 205 [M] · Original PDF p. 11

Answer. Threshold energy is the minimum collision energy required to reach a reactive state and form products; in the simple model it is associated with the activation barrier.

Question 206 [H] · Original PDF p. 11

Answer. The Boltzmann fraction of molecules with energy at least E_a depends exponentially on $-E_a/RT$. A larger E_a makes that fraction especially responsive to a temperature increase.

Question 207 [M] · Original PDF p. 11

Answer. Yes. A negative order can occur when a reactant inhibits the reaction, blocks catalyst sites, or changes a pre-equilibrium; it is an empirical rate-law effect, not a negative molecularity.

Question 208 [H] · Original PDF p. 12

Answer. Product binding can inhibit active sites, while adsorption can saturate or block a surface. Rates can then rise less than proportionally or even fall as concentration increases, producing zero or negative apparent orders.

Question 209 [M] · Original PDF p. 12

Answer. Rate is the concentration change per time and usually varies as concentrations change. k is the rate-law proportionality constant at fixed temperature and mechanism, independent of reactant concentrations.

Question 210 [H] · Original PDF p. 12

Answer. rate=k[A]. Doubling B has no effect because B has zero order in the stated rate law.

Question 211 [M] · Original PDF p. 12

Answer. Rate always has concentration/time units. Since rate=k(concentration)^n, k must have units concentration^(1−n) $time^{-1}$ so the equation is dimensionally consistent.

Question 212 [H] · Original PDF p. 12

Answer. 90% complete leaves 10%; t=ln(10)/k=2.3026/0.0231=99.7 min (about 100 min).

Question 213 [M] · Original PDF p. 12

Answer. Effective collision energy is the component of relative molecular energy available along the reaction coordinate; it must meet or exceed the threshold for reaction.

Question 214 [H] · Original PDF p. 12

Answer. E_a is the forward/reverse barrier from a reactant state to the transition state and is non-negative for a pathway. ΔH is the energy difference between products and reactants and may be positive or negative; they are not generally equal.

Question 215 [M] · Original PDF p. 12

Answer. Catalysts can bind selected reactants or transition states and provide a lower-energy pathway only for particular structures, so they may accelerate some reactions much more than others.

Question 216 [H] · Original PDF p. 12

Answer. For first order, 99% completion leaves $0.01[A]_0$, so $t_{99}=\ln 100/k$. Since $t_1/2=\ln 2/k$, $t_{99}/t_1/2=\ln 100/\ln 2=6.64$ half-lives.

UNIT 4: D- AND F-BLOCK ELEMENTS

Questions 217–288 · Original PDF pages 13–16

Question 217 [M] · Original PDF p. 13

Answer. A transition element is an element whose atom has an incomplete d subshell or which forms at least one common cation with an incomplete d subshell.

Question 218 [M] · Original PDF p. 13

Answer. General d-block configuration: $(n-1)d^{1–10}\ ns^{0–2}$; exceptions occur because the d and s subshell energies are close.

Question 219 [H] · Original PDF p. 13

Answer. Zn, Cd and Hg atoms and their common M^{2+} ions have filled d^{10} subshells, not partially filled d subshells; they therefore fail the strict transition-element definition.

Question 220 [M] · Original PDF p. 13

Answer. The ns and (n−1)d orbital energies are close, so different numbers of these electrons can participate in bonding or be removed, giving multiple oxidation states.

Question 221 [H] · Original PDF p. 13

Answer. The two outer ns electrons are usually lost first, giving M^{2+}. The (n−1)d electrons are more tightly held, so +2 is common for first-row metals.

Question 222 [M] · Original PDF p. 13

Answer. Partly filled d subshells have split d-orbital energy levels in a ligand field. Absorption of visible light promotes d–d transitions; remaining transmitted/reflected light gives colour.

Question 223 [H] · Original PDF p. 13

Answer. Ligand fields split the metal d orbitals. A photon with energy matching the splitting promotes an electron between them; selective visible absorption determines observed colour.

Question 224 [M] · Original PDF p. 13

Answer. A d^0 ion has no d electron to excite and a d^{10} ion has no vacant d state within the split d set; thus d–d transitions are unavailable. Charge-transfer bands can still colour some compounds.

Question 225 [H] · Original PDF p. 13

Answer. Changing oxidation state changes d-electron count and splitting; changing ligand alters the ligand-field strength and Δ. Both change transition energies and the wavelengths absorbed.

Question 226 [M] · Original PDF p. 13

Answer. Transition metals can access multiple oxidation states, adsorb reactants on surfaces and form intermediates, providing alternative pathways with lower activation barriers.

Question 227 [H] · Original PDF p. 13

Answer. Variable oxidation states allow electron transfer between reactants and metal centres. Adsorption brings reactants together and weakens bonds at the surface, lowering activation barriers.

Question 228 [M] · Original PDF p. 13

Answer. Unpaired d electrons have magnetic moments; their alignment with an applied field causes paramagnetism. Ions with all electrons paired are diamagnetic.

Question 229 [H] · Original PDF p. 13

Answer. Spin-only moment $\mu=\sqrt{[n(n+2)]}$ BM. For $n=3$, $\mu=\sqrt{15}=3.87$ BM.

Question 230 [M] · Original PDF p. 13

Answer. Many transition metals have similar atomic sizes and crystal structures, so atoms can replace one another in metallic lattices to form substitutional alloys.

Question 231 [M] · Original PDF p. 13

Answer. Interstitial compounds form when small atoms such as H, C or N occupy holes in a metal lattice. They are often hard/high-melting and may be non-stoichiometric.

Question 232 [H] · Original PDF p. 13

Answer. Small atoms fit into metal-lattice interstices and strongly distort/strengthen the lattice, restricting metal-atom motion; this often gives high hardness and melting points.

Question 233 [M] · Original PDF p. 13

Answer. Across the first transition series, atomic radii decrease modestly at first, vary little through the middle, and may rise slightly near the end; added d electrons partly shield increasing nuclear charge.

Question 234 [H] · Original PDF p. 13

Answer. Increasing nuclear charge, imperfect d-electron shielding, electron–electron repulsion and irregular stability of d configurations compete, producing a non-smooth ionisation-enthalpy trend.

Question 235 [M] · Original PDF p. 13

Answer. Lanthanoid contraction is the gradual decrease in atomic and especially Ln^{3+} ionic radii from La to Lu as nuclear charge increases across the series.

Question 236 [H] · Original PDF p. 13

Answer. Added 4f electrons shield the increasing nuclear charge poorly. Effective nuclear charge therefore rises across the series and contracts the electron cloud.

Question 237 [M] · Original PDF p. 13

Answer. Lanthanoid contraction makes Zr and Hf nearly equal in size, so they have very similar chemistry and are difficult to separate; it also affects hydroxide basicity and other size-dependent properties.

Question 238 [H] · Original PDF p. 14

Answer. The 4f electrons inserted between La and Lu shield poorly, causing lanthanoid contraction. Consequently Hf, after the lanthanoids, has almost the same radius as Zr above it.

Question 239 [M] · Original PDF p. 14

Answer. General lanthanoid configuration: $[Xe]4f^{0–14}5d^{0–1}6s^{2}$; most Ln atoms have 4f electrons, with limited 5d participation.

Question 240 [M] · Original PDF p. 14

Answer. Lanthanoids readily lose two 6s electrons and one 4f/5d electron to form Ln^{3+}. This oxidation state is common because the resulting ions are generally stable in compounds.

Question 241 [H] · Original PDF p. 14

Answer. +2 or +4 can provide especially stable empty, half-filled or filled 4f subshells, e.g. Ce^{4+} ($4f^{0}$), Eu^{2+} ($4f^{7}$), Tb^{4+} ($4f^{7}$), Yb^{2+} ($4f^{14}$); their stability also depends on lattice/hydration energies.

Question 242 [M] · Original PDF p. 14

Answer. General actinoid configuration: $[Rn]5f^{0–14}6d^{0–2}7s^{2}$, with substantial variation among elements.

Question 243 [H] · Original PDF p. 14

Answer. Actinoid 5f, 6d and 7s orbitals are closer in energy and 5f electrons are more available for bonding than 4f electrons in lanthanoids; several electron counts can therefore participate.

Question 244 [M] · Original PDF p. 14

Answer. Actinoids fill 5f orbitals whereas lanthanoids fill 4f; actinoids show a wider range of oxidation states and are all radioactive, while many lanthanoids have stable isotopes and predominantly +3 chemistry.

Question 245 [H] · Original PDF p. 14

Answer. Actinoids have very heavy nuclei with large proton–proton repulsion; their nuclei are unstable and undergo radioactive decay toward more stable configurations. All actinoids are radioactive, though half-lives vary greatly.

Question 246 [M] · Original PDF p. 14

Answer. Aqueous potassium dichromate is orange; chromium is in oxidation state +6.

Question 247 [M] · Original PDF p. 14

Answer. Permanganate ion MnO_4^- is deep purple/violet in aqueous solution.

Question 248 [H] · Original PDF p. 14

Answer. $Cr_2O_7^{2-}+14H^++6e^-\rightarrow 2Cr^{3+}+7H_2O$.

Question 249 [H] · Original PDF p. 14

Answer. $MnO_4^-+8H^++5e^-\rightarrow Mn^{2+}+4H_2O$.

Question 250 [M] · Original PDF p. 14

Answer. Acidified $KMnO_4$ accepts electrons and oxidises many reductants (e.g. $Fe^{2+}\rightarrow Fe^{3+}$ or $I^-\rightarrow I_2$); MnO_4^- is reduced mainly to pale Mn^{2+} in sufficiently acidic solution.

Question 251 [H] · Original PDF p. 14

Answer. H^+ participates in the reduction half-reaction, so its concentration shifts the redox equilibrium and changes the reduction potential/products. Acid favours Mn^{2+}; neutral/weakly alkaline conditions commonly yield MnO_2; strongly alkaline conditions can yield MnO_4^{2-}.

Question 252 [M] · Original PDF p. 14

Answer. In $K_2Cr_2O_7$: $2(+1)+2x+7(-2)=0$, so $x=+6$ for chromium.

Question 253 [M] · Original PDF p. 14

Answer. In $KMnO_4$: $(+1)+x+4(-2)=0$, so manganese is +7.

Question 254 [H] · Original PDF p. 14

Answer. For MnO_4^{2-}, $x+4(-2)=-2$, so Mn=+6. In MnO_4^-, Mn=+7; manganate is one oxidation state lower.

Question 255 [M] · Original PDF p. 14

Answer. Cr^{2+} (d^4) readily loses an electron to form particularly stable Cr^{3+} (d^3), so it reduces other species. Mn^{3+} (d^4) readily gains an electron to form half-filled Mn^{2+} (d^5), so it can oxidise other species.

Question 256 [H] · Original PDF p. 14

Answer. Cr^{3+} is d^3, with a stable t_2g^3 arrangement in an octahedral field; Mn^{2+} is d^5 and gains half-filled-subshell stability. Their stability also depends on environment and hydration.

Question 257 [M] · Original PDF p. 14

Answer. Several d and s electrons can contribute to metal–metal bonding; strong cohesive bonding requires substantial energy to separate atoms, giving high atomisation enthalpies.

Question 258 [H] · Original PDF p. 14

Answer. Melting points generally rise toward the middle of the first transition series, where metallic bonding is strong, then fall toward the end as d participation/cohesion declines. Subshell configurations, structure and bonding make the trend irregular.

Question 259 [M] · Original PDF p. 14

Answer. Disproportionation is simultaneous oxidation and reduction of the same oxidation state, e.g. $2Cu^+(aq)\rightarrow Cu^{2+}(aq)+Cu(s)$.

Question 260 [H] · Original PDF p. 14

Answer. In water, Cu^{2+} is strongly stabilised by hydration relative to Cu^+; the free-energy balance makes $2Cu^+\rightarrow Cu^{2+}+Cu$ favourable under standard aqueous conditions.

Question 261 [M] · Original PDF p. 14

Answer. Many transition-metal ions have partly filled d levels with ligand-field or charge-transfer absorptions in visible light; selective absorption produces vivid colours useful in pigments.

Question 262 [H] · Original PDF p. 15

Answer. High-spin d^5 has five unpaired electrons, $\mu=\sqrt{35}=5.92$ BM, and is strongly paramagnetic. Low-spin d^6 has t_2g^6, no unpaired electrons, $\mu\approx0$, and is diamagnetic (octahedral spin-only model).

Question 263 [M] · Original PDF p. 15

Answer. Lanthanoid contraction is due to increasing nuclear charge poorly shielded by 4f electrons; actinoid contraction arises similarly from poor 5f shielding. Both shrink trivalent ionic radii across their series; actinoid behaviour is less regular because 5f participation is greater.

Question 264 [H] · Original PDF p. 15

Answer. Ln^{3+} ions have similar charge and only gradually changing radii/chemistry, making their separation difficult. The small systematic contraction does produce gradual differences in basicity, complexation and related properties that can be exploited.

Question 265 [M] · Original PDF p. 15

Answer. Chromium has 3d and 4s valence electrons available, and strongly electronegative oxygen stabilises high oxidation state +6 in chromate/dichromate oxyanions.

Question 266 [H] · Original PDF p. 15

Answer. $2CrO_4^{2-}+2H^+\rightleftharpoons Cr_2O_7^{2-}+H_2O$. Acid shifts yellow chromate toward orange dichromate; added base removes H^+ and shifts back toward chromate.

Question 267 [M] · Original PDF p. 15

Answer. Cr: $[Ar]3d^54s^1$ rather than $3d^44s^2$. The 3d and 4s energies are close, and the half-filled d^5 subshell provides extra stability.

Question 268 [H] · Original PDF p. 15

Answer. Cr is $[Ar]3d^54s^1$ and Cu is $[Ar]3d^{10}4s^1$ rather than the simple expected distributions; near-degenerate 4s/3d levels allow configurations with half-filled or filled d subshells.

Question 269 [M] · Original PDF p. 15

Answer. Mn^{2+} is $[Ar]3d^5$, a particularly stable half-filled configuration. Removing a third electron breaks that arrangement, so the third ionisation enthalpy is relatively high.

Question 270 [H] · Original PDF p. 15

Answer. Fe^{2+} is d^6; Fe^{3+} is d^5, a half-filled subshell, which often favours Fe^{3+}. Relative stability is environment-dependent because hydration, lattice energy and ligand fields also contribute.

Question 271 [M] · Original PDF p. 15

Answer. Sc commonly forms Sc^{3+} by losing its two 4s and one 3d electron, leaving the stable [Ar] configuration.

Question 272 [H] · Original PDF p. 15

Answer. Early transition metals have fewer d electrons and can use more valence electrons in bonding, making high oxidation states more accessible. Later metals have higher ionisation costs and increasingly stable lower d configurations.

Question 273 [M] · Original PDF p. 15

Answer. Transition-metal ions are often small with high charge density and vacant/available orbitals; they can accept ligand lone pairs and form coordinate bonds.

Question 274 [H] · Original PDF p. 15

Answer. The number and arrangement of unpaired d electrons determine spin-only moment $\mu=\sqrt{[n(n+2)]}$ BM. Any unpaired electrons give paramagnetism; all paired gives diamagnetism; ligand strength can change pairing in complexes.

Question 275 [M] · Original PDF p. 15

Answer. Cu^{2+} is d^9 with a partially filled d level, allowing visible d–d transitions. Zn^{2+} is d^{10} with no d–d transition and is usually colourless unless charge-transfer absorption occurs.

Question 276 [H] · Original PDF p. 15

Answer. Ligand identity sets ligand-field strength through its bonding and interaction with metal d orbitals, changing Δ. The absorbed photon energy and d–d transition wavelength therefore change.

Question 277 [M] · Original PDF p. 15

Answer. Basicity of lanthanoid hydroxides generally decreases from $La(OH)_3$ to $Lu(OH)_3$ as Ln^{3+} radius contracts and polarising power/covalent character increase.

Question 278 [H] · Original PDF p. 15

Answer. Across La to Lu, Ln^{3+} size decreases while charge remains +3. Greater polarisation strengthens Ln–O covalent character and reduces hydroxide basicity; $La(OH)_3$ is more basic than $Lu(OH)_3$.

Question 279 [M] · Original PDF p. 15

Answer. Actinoid 5f orbitals are more spatially extended and can participate in bonding, and actinoids commonly access higher oxidation states and strong metal–ligand interactions; this favours complex formation.

Question 280 [H] · Original PDF p. 15

Answer. The 5f, 6d and 7s levels are close in energy, so several configurations and oxidation states are accessible. This, plus greater 5f bonding participation and radioactivity, makes actinoid chemistry complex.

Question 281 [M] · Original PDF p. 16

Answer. Chromite ore is oxidised/roasted to soluble chromate. H_2SO_4 acidifies the chromate solution, shifting $2CrO_4^{2-}+2H^+ \rightleftharpoons Cr_2O_7^{2-}+H_2O$ toward dichromate; adding KCl then crystallises less-soluble $K_2Cr_2O_7$.

Question 282 [H] · Original PDF p. 16

Answer. $MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$.

Question 283 [M] · Original PDF p. 16

Answer. In acid, H^+ drives the MnO_4^-/Mn^{2+} reduction half-reaction, whose potential is more positive; permanganate is consequently a stronger oxidant than when reduced to MnO_2 in neutral medium.

Question 284 [H] · Original PDF p. 16

Answer. $K_2Cr_2O_7$ is available in high purity, stable in air, non-hygroscopic, has a relatively high molar mass and can be weighed accurately to prepare a standard solution.

Question 285 [M] · Original PDF p. 16

Answer. Delocalised mobile electrons carry charge and transport thermal energy through the metallic lattice, giving transition metals generally good electrical and thermal conductivity.

Question 286 [H] · Original PDF p. 16

Answer. Zn and Cd have filled $(n-1)d^{10}$ subshells and fewer unpaired d electrons to strengthen metal–metal cohesion; their metallic bonding and atomisation enthalpies are comparatively lower.

Question 287 [M] · Original PDF p. 16

Answer. Transition metals involve filling d orbitals in the outer transition series; inner-transition metals involve progressive filling of inner f orbitals (lanthanoids 4f, actinoids 5f).

Question 288 [H] · Original PDF p. 16

Answer. E° reflects several energetic terms—atomisation, ionisation, hydration/solvation and entropy—not just the electron configuration. Their balance depends on the species and medium, so configuration alone is insufficient.

UNIT 5: COORDINATION COMPOUNDS

Questions 289–358 · Original PDF pages 17–20

Question 289 [M] · Original PDF p. 17

Answer. A coordination compound contains a central metal atom/ion bonded to surrounding ligands by coordinate bonds; e.g. $[Co(NH_3)_6]Cl_3$.

Question 290 [M] · Original PDF p. 17

Answer. A ligand donates one or more lone pairs to a central metal. By denticity, ligands are monodentate (one donor atom), bidentate (two) or polydentate (several); e.g. NH_3, en and EDTA.

Question 291 [M] · Original PDF p. 17

Answer. Coordination number is the number of ligand donor atoms directly bonded to the metal. In $[Co(NH_3)_6]^{3+}$ it is 6.

Question 292 [H] · Original PDF p. 17

Answer. In $[Co(NH_3)_5Cl]Cl_2$ the complex cation has charge +2. With NH_3 neutral and coordinated Cl^- at −1, $x-1=+2$, so Co is +3.

Question 293 [M] · Original PDF p. 17

Answer. A double salt dissociates in water into all constituent simple ions and loses its identity. A coordination compound retains complex ions in solution, e.g. $[Fe(CN)_6]^{4-}$.

Question 294 [M] · Original PDF p. 17

Answer. Werner’s primary valency is ionisable and corresponds mainly to oxidation state; secondary valency is coordination number, satisfied by ligands and directional in space.

Question 295 [H] · Original PDF p. 17

Answer. $CoCl_3{\cdot}6NH_3$ is $[Co(NH_3)_6]Cl_3$: six secondary valencies and three ionisable Cl^-. $CoCl_3{\cdot}5NH_3$ is $[Co(NH_3)_5Cl]Cl_2$: five NH_3 plus one coordinated Cl^- give coordination number 6, with two ionisable Cl^-.

Question 296 [M] · Original PDF p. 17

Answer. Hexaamminecobalt(III) chloride.

Question 297 [M] · Original PDF p. 17

Answer. Potassium hexacyanidoferrate(II).

Question 298 [H] · Original PDF p. 17

Answer. $[Co(NH_3)_5Cl]Cl_2$.

Question 299 [M] · Original PDF p. 17

Answer. A chelating ligand binds a metal through two or more donor atoms, forming one or more rings; ethane-1,2-diamine (en) or oxalate is an example.

Question 300 [H] · Original PDF p. 17

Answer. Multidentate ligands form several metal–donor bonds at once, and chelation often has a favourable entropy because several coordinated monodentate molecules are released as one multidentate ligand binds. Ring formation and rebinding make dissociation less favourable than for analogous monodentate ligands.

Question 301 [M] · Original PDF p. 17

Answer. Homoleptic complexes contain only one kind of ligand; heteroleptic complexes contain two or more different ligand types.

Question 302 [M] · Original PDF p. 17

Answer. In VBT, ligands donate lone pairs into vacant hybrid orbitals on the metal. Hybrid-orbital arrangement predicts geometry; the metal’s unpaired d electrons help predict magnetic behaviour.

Question 303 [H] · Original PDF p. 17

Answer. In $[Ni(CN)_4]^{2-}$, Ni^{2+} is d^8 and CN^- is strong-field; d electrons pair, giving dsp^2 hybridisation, square-planar geometry and diamagnetism.

Question 304 [H] · Original PDF p. 17

Answer. In $[NiCl_4]^{2-}$, Ni^{2+} is d^8 and Cl^- weak-field; the complex is sp^3 tetrahedral with two unpaired electrons and is paramagnetic.

Question 305 [M] · Original PDF p. 17

Answer. Crystal-field splitting is removal of degeneracy of metal d orbitals by the electrostatic field of surrounding ligands; the energy separation depends on geometry and ligand strength.

Question 306 [M] · Original PDF p. 17

Answer. Strong-field ligands produce a large splitting Δ and may cause electron pairing; weak-field ligands give a smaller Δ, often leaving electrons unpaired in high-spin complexes.

Question 307 [H] · Original PDF p. 17

Answer. Both contain Fe^{2+} (d^6). CN^- gives $\Delta_o > P$, so $[Fe(CN)_6]^{4-}$ is low-spin t_{2g}^6; weak-field F^- gives $\Delta_o < P$, so $[FeF_6]^{4-}$ is high-spin $t_{2g}^4 e_g^2$.

Question 308 [M] · Original PDF p. 17

Answer. In an octahedral field, the five d orbitals split into lower-energy t_{2g} (d_{xy},d_{xz},d_{yz}) and higher-energy e_g (d_{z^2},$d_{x^2-y^2}$), separated by Δ_o.

Question 309 [M] · Original PDF p. 17

Answer. In a tetrahedral field, e (d_{z^2},$d_{x^2-y^2}$) is lower and t_2 (d_{xy},d_{xz},d_{yz}) higher; the order is opposite to octahedral and the gap is Δ_t.

Question 310 [H] · Original PDF p. 17

Answer. In tetrahedral geometry only four ligands approach and none lies directly on the axes; interaction with d lobes is smaller. Consequently $\Delta_t \approx (4/9)\Delta_o$ for the same metal/ligand (approximately).

Question 311 [M] · Original PDF p. 17

Answer. Crystal-field stabilisation energy is the net lowering in d-electron energy relative to the unsplit barycentre due to occupation of the lower-energy ligand-field orbitals.

Question 312 [H] · Original PDF p. 17

Answer. Octahedral d^3 is $t_{2g}^3 e_g^0$. $CFSE = 3(-0.4\Delta_o) = -1.2\Delta_o$, i.e. stabilisation of $1.2\Delta_o$; pairing energy is excluded.

Question 313 [M] · Original PDF p. 17

Answer. Partly filled d orbitals can undergo ligand-field d–d transitions by absorbing visible photons, and charge-transfer transitions may also absorb light; unabsorbed light gives observed colour.

Question 314 [H] · Original PDF p. 18

Answer. A photon is absorbed when hc/λ matches the ligand-field transition energy (approximately Δ). Thus $\lambda = hc/\Delta$; larger splitting corresponds to shorter absorbed wavelength.

Question 315 [M] · Original PDF p. 18

Answer. Magnetic susceptibility/moment estimates unpaired-electron count. High-spin complexes usually have more unpaired electrons and larger paramagnetic moments than their low-spin counterparts; all-paired complexes are diamagnetic.

Question 316 [H] · Original PDF p. 18

Answer. Spin-only $\mu = \sqrt{[n(n+2)]}$ BM $= \sqrt{[4\times6]} = \sqrt{24} = 4.90$ BM.

Question 317 [M] · Original PDF p. 18

Answer. Geometrical isomerism occurs when the same ligands occupy different relative positions around the metal, e.g. cis and trans arrangements, without changing connectivity.

Question 318 [M] · Original PDF p. 18

Answer. Optical isomers are non-superimposable mirror-image coordination structures (enantiomers) that rotate plane-polarised light in opposite directions.

Question 319 [H] · Original PDF p. 18

Answer. Square-planar $[MA_2B_2]$ has cis isomer with identical A ligands adjacent and trans isomer with A ligands opposite; these are distinct spatial arrangements.

Question 320 [H] · Original PDF p. 18

Answer. Two: cis-$[MA_4B_2]$ and trans-$[MA_4B_2]$, according to the relative positions of the two B ligands.

Question 321 [M] · Original PDF p. 18

Answer. $[Co(NH_3)_5(NO_2)]Cl_2$ can bind NO_2^- through N (nitro) or O (nitrito), giving linkage isomers.

Question 322 [M] · Original PDF p. 18

Answer. Ionisation isomers exchange a ligand inside the coordination sphere with a counter-ion outside, producing different ions in solution; e.g. $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$.

Question 323 [H] · Original PDF p. 18

Answer. Linkage: $[Co(NH_3)_5(NO_2)]Cl_2$ (nitro/nitrito donor atom). Ionisation: $[Co(NH_3)_5Br]SO_4$ vs $[Co(NH_3)_5SO_4]Br$. Hydrate: $[Cr(H_2O)_6]Cl_3$ vs $[Cr(H_2O)_5Cl]Cl_2\cdot H_2O$.

Question 324 [M] · Original PDF p. 18

Answer. Complex formation gives characteristic colours or selective precipitates and can mask interfering ions, allowing ions to be separated or identified in qualitative analysis.

Question 325 [M] · Original PDF p. 18

Answer. In cyanide leaching, Au/Ag forms soluble dicyano complexes (e.g. $4Au+8CN^-+O_2+2H_2O\rightarrow 4[Au(CN)_2]^-+4OH^-$); metal is then recovered from solution.

Question 326 [M] · Original PDF p. 18

Answer. Haemoglobin's Fe-containing heme coordination sites bind O_2 reversibly, enabling oxygen transport in blood; binding must remain reversible and not permanently oxidise the iron.

Question 327 [H] · Original PDF p. 18

Answer. Chlorophyll has central Mg^{2+} in a porphyrin-like coordination framework for light capture in photosynthesis. Vitamin B_{12} (cobalamin) contains Co coordinated within a corrin ring and is essential in several enzyme-mediated reactions.

Question 328 [M] · Original PDF p. 18

Answer. EDTA is hexadentate and can bind through six donor atoms, forming multiple bonds and chelate rings around many metal ions; this yields stable complexes and strong sequestration.

Question 329 [H] · Original PDF p. 18

Answer. Inner-orbital octahedral complexes use two $(n-1)d$, one ns and three np orbitals (d^2sp^3), often low spin. Outer-orbital complexes use ns, np and nd (sp^3d^2), often high spin; this is VBT's model.

Question 330 [M] · Original PDF p. 18

Answer. The spectrochemical series orders ligands by increasing ability to split metal d orbitals, e.g. $I^-<Br^-<Cl^-<F^-<OH^-<H_2O<NH_3<en<CN^-<CO$.

Question 331 [H] · Original PDF p. 18

Answer. If $\Delta>P$, placing an electron in a higher orbital costs more than pairing in a lower one, so pairing occurs. If $\Delta<P$, occupation of separate higher orbitals costs less, favouring high spin.

Question 332 [M] · Original PDF p. 18

Answer. For $[Fe(CN)_6]^{3-}$, $x+6(-1)=-3$ gives Fe=+3. Six monodentate CN^- donor atoms give coordination number 6.

Question 333 [H] · Original PDF p. 18

Answer. Mn in $[Mn(CN)_6]^{3-}$ is +3, d^4. Strong-field CN^- gives low-spin $t_{2g}^4e_g^0$, with two unpaired electrons.

Question 334 [M] · Original PDF p. 18

Answer. EAN=metal atomic number−oxidation state+electrons donated by ligands; stability is sometimes associated with reaching a noble-gas electron count, but it is only an empirical guideline.

Question 335 [H] · Original PDF p. 18

Answer. Many stable complexes do not attain noble-gas EAN; ligand field, covalency, steric effects, charge, kinetics and thermodynamics can dominate. The rule is not a universal stability criterion.

Question 336 [M] · Original PDF p. 19

Answer. A monodentate ligand binds through one donor atom; a bidentate ligand binds through two donor atoms to the same metal, often forming a chelate ring.

Question 337 [H] · Original PDF p. 19

Answer. Square-planar Pt(II) complexes of formula $[Pt(NH_3)_2Cl_2]$ have cis (like ligands adjacent) and trans (opposite) arrangements. Cisplatin is the cis form and is an important anticancer drug; the trans form is not equivalently effective.

Question 338 [M] · Original PDF p. 19

Answer. en is neutral and two Cl^- contribute −2. For complex charge +1, x−2=+1, so Co=+3. Two en ligands donate four atoms and two Cl^- donate two: coordination number 6.

Question 339 [H] · Original PDF p. 19

Answer. Tetraaquadichloridochromium(III) chloride.

Question 340 [M] · Original PDF p. 19

Answer. Ethane-1,2-diamine (en) is bidentate, denticity 2, donating through its two nitrogen atoms.

Question 341 [H] · Original PDF p. 19

Answer. Fe in $[Fe(CN)_6]^{3-}$ is Fe^{3+}, d^5. Under strong-field conditions it is low-spin t_{2g}^5 with one unpaired electron; VBT predicts inner-orbital d^2sp^3 octahedral geometry.

Question 342 [M] · Original PDF p. 19

Answer. Zn^{2+} is d^{10}, so all metal d electrons are paired; $[Zn(NH_3)_4]^{2+}$ is therefore diamagnetic in the usual description.

Question 343 [H] · Original PDF p. 19

Answer. Octahedral d^6 high spin $t_{2g}^4e_g^2$ has four unpaired electrons, μ≈4.90 BM; low spin t_{2g}^6 has none and is diamagnetic (spin-only values).

Question 344 [M] · Original PDF p. 19

Answer. Δ is the energy gap between ligand-field orbital sets. Pairing energy P is the energetic cost of placing two electrons in one orbital; comparing Δ and P determines spin state.

Question 345 [H] · Original PDF p. 19

Answer. Octahedral d^4 high spin: $t_{2g}^3e_g^1$, four unpaired. Low spin: $t_{2g}^4e_g^0$, two unpaired; low spin is favoured when Δ_o>P.

Question 346 [M] · Original PDF p. 19

Answer. $[Co(en)_3]^{3+}$ is an example; its Δ and Λ forms are non-superimposable mirror images.

Question 347 [H] · Original PDF p. 19

Answer. In cis-$[M(en)_2Cl_2]^+$, the two chelate rings can form a chiral arrangement with Δ/Λ mirror-image forms and no improper symmetry element. The trans arrangement has symmetry that makes it achiral in the usual octahedral case.

Question 348 [M] · Original PDF p. 19

Answer. The coordination sphere is the central metal together with directly attached ligands, conventionally written inside square brackets.

Question 349 [H] · Original PDF p. 19

Answer. $[Co(NH_3)_5Cl]Cl_2 \rightarrow [Co(NH_3)_5Cl]^{2+} + 2Cl^-$, giving three ions per formula unit if the complex ion remains intact.

Question 350 [M] · Original PDF p. 19

Answer. Geometry reflects coordination number, ligand size/repulsion, metal electronic structure and ligand-field stabilisation; these favour characteristic arrangements such as tetrahedral, square planar or octahedral.

Question 351 [H] · Original PDF p. 19

Answer. VBT gives qualitative hybridisation/geometry and pairing but does not quantitatively explain d-orbital splitting, colours, absorption spectra, spectrochemical ordering or many magnetic trends.

Question 352 [M] · Original PDF p. 19

Answer. The chelate effect is the enhanced thermodynamic stability often observed when a multidentate ligand replaces comparable monodentate ligands, commonly due to favourable entropy and multiple simultaneous bonds.

Question 353 [H] · Original PDF p. 19

Answer. A masking ligand binds an interfering metal ion strongly, lowering its free-ion concentration and preventing its usual precipitation or colour reaction while other ions are tested.

Question 354 [M] · Original PDF p. 19

Answer. $[Ag(NH_3)_2]^+$ is the soluble diamminesilver(I) complex in Tollens' reagent, used to identify aldehydes by their reduction of Ag(I) to metallic silver.

Question 355 [H] · Original PDF p. 19

Answer. A complexing ligand removes free metal ions from solution. This shifts the dissolution equilibrium of a sparingly soluble salt toward more dissolved ions, increasing apparent solubility.

Question 356 [M] · Original PDF p. 19

Answer. $[Cu(NH_3)_4]SO_4$.

Question 357 [H] · Original PDF p. 20

Answer. In octahedral $[MA_3B_3]$, fac has the three A ligands on one face, all mutually cis. Mer has the three A ligands along a meridian, with one A–A pair trans.

Question 358 [H] · Original PDF p. 20

Answer. Octahedral d^5 high-spin $t_{2g}^3 e_g^2$ has five unpaired electrons; low-spin $t_{2g}^5 e_g^0$ has one unpaired electron.

UNIT 6: HALOALKANES AND HALOARENES

Questions 359–428 · Original PDF pages 21–24

Question 359 [M] · Original PDF p. 21

Answer. A haloalkane has halogen bonded to an aliphatic sp^3 carbon (e.g. CH_3CH_2Cl). A haloarene has halogen directly bonded to an aromatic-ring carbon (e.g. chlorobenzene).

Question 360 [M] · Original PDF p. 21

Answer. 2-Bromobutane.

Question 361 [M] · Original PDF p. 21

Answer. Halogens are more electronegative than carbon, so the C–X bond is polar: $C\delta^+–X\delta^-$.

Question 362 [H] · Original PDF p. 21

Answer. Bond strength decreases C–F>C–Cl>C–Br>C–I as bond length increases. For substitution, easier C–X cleavage generally gives RI>RBr>RCl≫RF, although mechanism and substrate also matter.

Question 363 [M] · Original PDF p. 21

Answer. Nucleophilic substitution replaces a leaving group (often halide) on carbon with a nucleophile: $R–X+Nu^-\rightarrow R–Nu+X^-$.

Question 364 [M] · Original PDF p. 21

Answer. SN1 is two-step, through a carbocation; rate=k[RX], often favoured by tertiary substrate/polar protic solvent and gives racemisation tendency. SN2 is concerted backside attack; rate=k[RX][Nu^-], favoured by less hindered substrate/strong nucleophile and gives inversion.

Question 365 [H] · Original PDF p. 21

Answer. Tertiary substrates form relatively stable carbocations and are sterically hindered to backside attack, favouring SN1. Primary substrates form unstable carbocations but allow backside approach, favouring SN2; solvent/nucleophile can alter the outcome.

Question 366 [M] · Original PDF p. 21

Answer. Racemisation is loss of optical purity as an initially optically active substrate forms a mixture of enantiomers, often through attack on both faces of a planar carbocation.

Question 367 [H] · Original PDF p. 21

Answer. SN2 proceeds by backside attack as the C–Nu bond forms and C–X breaks, causing inversion of configuration at the reacting stereogenic centre (Walden inversion).

Question 368 [M] · Original PDF p. 21

Answer. Optical activity is the ability of a chiral substance to rotate the plane of plane-polarised light; enantiomers rotate it equally in opposite directions.

Question 369 [H] · Original PDF p. 21

Answer. SN1 forms a roughly planar carbocation, which can be attacked from either face, producing both configurations. Ion-pair effects often make racemisation incomplete rather than exactly 50:50.

Question 370 [M] · Original PDF p. 21

Answer. SN1 reactivity generally 3°>2°>1° because increasing alkyl substitution stabilises the rate-determining carbocation.

Question 371 [H] · Original PDF p. 21

Answer. Major product is butan-2-ol, $CH_3CH(OH)CH_2CH_3$, by nucleophilic substitution with aqueous OH^- replacing Br^-; secondary-substrate elimination can be a minor competing pathway depending on conditions.

Question 372 [M] · Original PDF p. 21

Answer. Dehydrohalogenation is elimination of HX from adjacent (α and β) carbons of a haloalkane to form an alkene, commonly with alcoholic base and heat.

Question 373 [H] · Original PDF p. 21

Answer. Aqueous KOH favours nucleophilic substitution to an alcohol. Alcoholic KOH with heat favours β-elimination to an alkene; substrate and conditions can allow competing pathways.

Question 374 [M] · Original PDF p. 21

Answer. Aryl C–X has partial double-bond character from resonance, the bonded carbon is sp^2, and backside attack/phenyl-cation formation is difficult; haloarenes therefore resist ordinary SN reactions.

Question 375 [H] · Original PDF p. 21

Answer. A chlorine lone pair overlaps with the benzene π system. Resonance contributors place π bonding between C and Cl, giving the aryl C–Cl bond partial double-bond character and shortening/strengthening it.

Question 376 [M] · Original PDF p. 21

Answer. Cl donates a lone pair by resonance, increasing electron density mainly at ortho and para positions and stabilising the corresponding arenium intermediates; these positions are therefore directed despite deactivation.

Question 377 [H] · Original PDF p. 21

Answer. The strong −I effect withdraws electron density and deactivates the ring. Lone-pair resonance donation stabilises ortho/para substitution intermediates, so halogens remain ortho/para directors.

Question 378 [M] · Original PDF p. 21

Answer. Nitration gives ortho- and para-nitrochlorobenzene, with the para isomer generally major because it is less sterically crowded.

Question 379 [M] · Original PDF p. 21

Answer. Wurtz reaction couples two alkyl halides with sodium in dry ether: $2R–X+2Na\rightarrow R–R+2NaX$.

Question 380 [M] · Original PDF p. 22

Answer. Fittig reaction couples aryl halides with sodium in dry ether: $2Ar–X+2Na\rightarrow Ar–Ar+2NaX$.

Question 381 [M] · Original PDF p. 22

Answer. Wurtz–Fittig reaction couples an aryl and an alkyl halide with sodium in dry ether: $Ar–X+R–X+2Na\rightarrow Ar–R+2NaX$.

Question 382 [H] · Original PDF p. 22

Answer. Principal Wurtz–Fittig product is toluene: $C_6H_5Br+CH_3Br+2Na\rightarrow C_6H_5CH_3+2NaBr$. Homocoupling side-products (biphenyl and ethane) may also form.

Question 383 [M] · Original PDF p. 22

Answer. Heat the haloalkane with aqueous NaOH or KOH to replace X by OH: $R–X+OH^-\rightarrow R–OH+X^-$.

Question 384 [H] · Original PDF p. 22

Answer. Convert ethanol to bromoethane with HBr (often generated from NaBr/KBr and concentrated H_2SO_4) or use PBr_3: $CH_3CH_2OH\rightarrow CH_3CH_2Br$.

Question 385 [M] · Original PDF p. 22

Answer. An allylic halide has X on an sp^3 carbon adjacent to C=C; a vinylic halide has X directly attached to an alkene (sp^2) carbon.

Question 386 [H] · Original PDF p. 22

Answer. Vinyl C–Cl has partial double-bond character through conjugation, is shorter/stronger at an sp^2 carbon and resists backside attack; ethyl chloride has an ordinary sp^3 C–Cl bond.

Question 387 [M] · Original PDF p. 22

Answer. Dichloromethane is used as an industrial solvent and in paint strippers. Emissions contribute to stratospheric ozone depletion despite its relatively short atmospheric lifetime (Australian Industrial Chemicals Introduction Scheme evaluation, 2021).

Question 388 [M] · Original PDF p. 22

Answer. Chloroform has been used as a solvent and historically as an anaesthetic. It can damage liver/kidneys and depress the central nervous system; it is unsafe for unsupervised exposure.

Question 389 [M] · Original PDF p. 22

Answer. CCl_4 is toxic and persistent, and chlorine radicals released in the stratosphere catalytically deplete ozone; its production/use has been heavily restricted.

Question 390 [M] · Original PDF p. 22

Answer. Iodoform is triiodomethane, CHI_3, a yellow crystalline solid historically used as an antiseptic/dressing because it releases iodine-containing species.

Question 391 [M] · Original PDF p. 22

Answer. Freons are chlorofluorocarbon (CFC) refrigerants/propellants. Stable CFCs reach the stratosphere and release chlorine radicals that deplete ozone; many were phased out under the Montreal Protocol.

Question 392 [M] · Original PDF p. 22

Answer. DDT is persistent, fat-soluble and bioaccumulates/biomagnifies, harming wildlife (notably birds) and potentially exposing people through food chains.

Question 393 [H] · Original PDF p. 22

Answer. DDT is notably persistent and bioaccumulative. Many newer pesticides are designed to degrade faster, but persistence and bioaccumulation vary by compound; some modern pesticides remain hazardous, so a blanket claim that all are safer is not justified.

Question 394 [M] · Original PDF p. 22

Answer. For the same alkyl group, molar mass and polarizability generally rise F→Cl→Br→I, strengthening dispersion forces; boiling points usually rise in that order (with structural exceptions).

Question 395 [H] · Original PDF p. 22

Answer. Branching makes isomers more compact, reducing intermolecular contact surface and London dispersion forces; less energy is required to separate molecules, so boiling point generally falls.

Question 396 [M] · Original PDF p. 22

Answer. Haloalkanes cannot hydrogen-bond effectively with water, and their hydrophobic alkyl group makes hydration energetically unfavourable; only slight amounts dissolve.

Question 397 [H] · Original PDF p. 22

Answer. Many organic solvents have compatible polarizability and dispersion interactions with haloalkanes, so solute–solvent attractions can offset disruption of solvent–solvent interactions despite C–X bond polarity.

Question 398 [M] · Original PDF p. 22

Answer. A Grignard reagent is an organomagnesium halide, RMgX, containing a strongly polar C–Mg bond and behaving as a powerful nucleophilic/basic carbon reagent.

Question 399 [H] · Original PDF p. 22

Answer. Water protonates the carbon end and destroys the reagent: $RMgX + H_2O \rightarrow RH + Mg(OH)X$. Dry ether coordinates/stabilises Mg and must also be anhydrous.

Question 400 [M] · Original PDF p. 22

Answer. $CH_3CH_2Br + Mg \rightarrow CH_3CH_2MgBr$ in dry ether.

Question 401 [H] · Original PDF p. 22

Answer. $RMgX+CO_2 \rightarrow RCO_2MgX$; acidic hydrolysis gives RCOOH. The carboxylic acid has one more carbon than R.

Question 402 [M] · Original PDF p. 22

Answer. Use sodium fusion (Lassaigne's test) to convert halogen to NaX in the extract; acidify with dilute HNO_3, then add $AgNO_3$. AgCl is white, AgBr pale yellow and AgI yellow; confirmatory solubility tests can distinguish them.

Question 403 [H] · Original PDF p. 23

Answer. Covalently bound halogen may not release X^- simply on contact with aqueous $AgNO_3$, especially for aryl/vinyl halides. The test depends on hydrolysis/ionisation and solubility, so sample preparation and suitable reaction conditions are needed.

Question 404 [M] · Original PDF p. 23

Answer. A nucleophile donates an electron pair to an electrophilic atom (often carbon) to form a bond. A base accepts a proton; many species can do both, but the attacked site differs.

Question 405 [H] · Original PDF p. 23

Answer. Polar protic solvents stabilise ions/carbocations and solvate anionic nucleophiles, often favouring SN1 and slowing SN2. Polar aprotic solvents solvate cations but leave nucleophiles reactive, often accelerating SN2.

Question 406 [M] · Original PDF p. 23

Answer. A leaving group departs from carbon with the bonding electron pair, enabling substitution. Stable, weakly basic anions generally leave more readily.

Question 407 [H] · Original PDF p. 23

Answer. Typical ability: $I^- > Br^- > Cl^- \gg F^-$. Larger, more polarizable anions are more stable and weaker bases, and the C–I bond is weakest; the exact rate still depends on substrate and mechanism.

Question 408 [M] · Original PDF p. 23

Answer. 2-Chlorobutane.

Question 409 [H] · Original PDF p. 23

Answer. Propene is the major product: $CH_3CHBrCH_3+KOH(alc) \rightarrow CH_3CH=CH_2+KBr+H_2O$. Alcoholic base and heat favour concerted E2 β-elimination.

Question 410 [M] · Original PDF p. 23

Answer. Tert-butyl chloride forms a relatively stable tertiary carbocation; a polar protic solvent stabilises the developing ions, accelerating the SN1 ionisation step.

Question 411 [H] · Original PDF p. 23

Answer. SN1 has two transition states separated by a carbocation intermediate; the first, ionisation barrier is often rate-determining. SN2 has one concerted transition state and one barrier, with simultaneous C–Nu formation and C–X cleavage.

Question 412 [M] · Original PDF p. 23

Answer. Inversion of configuration is reversal of the spatial arrangement at a stereogenic carbon, as in backside SN2 attack (Walden inversion).

Question 413 [H] · Original PDF p. 23

Answer. Likely SN2: the strong nucleophile attacks from the backside in polar aprotic solvent, producing inversion at the reacting chiral centre, provided steric hindrance does not prevent SN2.

Question 414 [M] · Original PDF p. 23

Answer. Chlorobenzene's aryl C–Cl bond has resonance partial double-bond character and is attached to an sp^2 carbon; ordinary backside substitution is difficult, so hydrolysis requires much harsher conditions than for alkyl chlorides.

Question 415 [H] · Original PDF p. 23

Answer. Ortho/para electron-withdrawing groups stabilise the negatively charged Meisenheimer addition intermediate by resonance delocalisation, lowering the barrier for nucleophilic aromatic substitution.

Question 416 [M] · Original PDF p. 23

Answer. Bromoethane hydrolyses to ethanol: $CH_3CH_2Br + OH^- \rightarrow CH_3CH_2OH + Br^-$.

Question 417 [H] · Original PDF p. 23

Answer. Add HBr to propene in the presence of peroxide (ROOR), usually with heat/light: radical anti-Markovnikov addition gives $CH_3CH_2CH_2Br$ (1-bromopropane).

Question 418 [M] · Original PDF p. 23

Answer. Dry ether solvates/stabilises the sodium and organometallic intermediates and provides an inert medium; moisture would consume sodium or protonate reactive species.

Question 419 [H] · Original PDF p. 23

Answer. Each haloalkane can couple with either partner, producing both cross-coupled and homocoupled products. The mixture is difficult to avoid unless the reactants are chosen carefully.

Question 420 [M] · Original PDF p. 23

Answer. UV light breaks C–Cl bonds in stratospheric CFCs, releasing chlorine radicals. These radicals catalytically convert ozone to oxygen and are regenerated.

Question 421 [H] · Original PDF p. 23

Answer. $Cl\cdot + O_3 \rightarrow ClO\cdot + O_2$; then $ClO\cdot + O\cdot \rightarrow Cl\cdot + O_2$. Chlorine radical is regenerated, so the net reaction is $O_3 + O \rightarrow 2O_2$ and one radical can destroy many ozone molecules.

Question 422 [M] · Original PDF p. 23

Answer. A geminal dihalide has both halogen atoms on the same carbon; a vicinal dihalide has them on adjacent carbons.

Question 423 [H] · Original PDF p. 23

Answer. Allyl chloride is highly reactive because its transition state/carbocation is resonance-stabilised; ethyl chloride has ordinary primary-halide reactivity; vinyl chloride is much less reactive because its sp^2 C–Cl bond has partial double-bond character. Typical order: allyl chloride>ethyl chloride≫vinyl chloride.

Question 424 [M] · Original PDF p. 24

Answer. Halogens are much heavier and more polarizable than hydrogen, so replacing H with one or more halogens raises molar mass and often density relative to the corresponding hydrocarbon.

Question 425 [H] · Original PDF p. 24

Answer. A new stereogenic centre forms if substitution gives a carbon attached to four different groups. Assign R/S by CIP priorities (atomic number, then successive ties); determine whether the mechanism gives inversion, racemisation or another stereochemical outcome separately.

Question 426 [M] · Original PDF p. 24

Answer. Propanenitrile (ethyl cyanide), $CH_3CH_2C{\equiv}N$; KCN attacks through carbon, adding one carbon to the chain.

Question 427 [H] · Original PDF p. 24

Answer. KCN is ionic and supplies free CN^-, which attacks through carbon to give propanenitrile, CH_3CH_2CN. Covalent AgCN favours nitrogen-end attack, giving ethyl isocyanide, CH_3CH_2NC.

Question 428 [H] · Original PDF p. 24

Answer. SN2 is a single concerted collision of haloalkane and nucleophile, so its rate law is $\text{rate} = k[RX][Nu^-]$; increasing either reactant concentration increases the frequency of productive encounters.

UNIT 7: ALCOHOLS, PHENOLS AND ETHERS

Questions 429–500 · Original PDF pages 25–28

Question 429 [M] · Original PDF p. 25

Answer. Alcohols have –OH on an aliphatic carbon (ethanol, CH_3CH_2OH); phenols have –OH directly on an aromatic ring (phenol, C_6H_5OH); ethers have oxygen between carbon groups (diethyl ether, $C_2H_5OC_2H_5$).

Question 430 [M] · Original PDF p. 25

Answer. Butan-2-ol.

Question 431 [M] · Original PDF p. 25

Answer. In a primary alcohol the carbon bearing OH is attached to one other carbon; in a secondary alcohol to two; in a tertiary alcohol to three.

Question 432 [M] · Original PDF p. 25

Answer. Alcohol molecules form intermolecular hydrogen bonds through –OH groups. More energy is needed to separate them than comparable alkanes, which rely mainly on dispersion forces.

Question 433 [H] · Original PDF p. 25

Answer. Intermolecular hydrogen bonds raise alcohol boiling points. The –OH group hydrogen-bonds with water, making lower alcohols soluble; as the hydrophobic alkyl chain grows, water solubility decreases.

Question 434 [M] · Original PDF p. 25

Answer. Hydrate ethene with steam over a phosphoric-acid catalyst at elevated temperature/pressure: $CH_2{=}CH_2+H_2O \rightleftharpoons CH_3CH_2OH$.

Question 435 [M] · Original PDF p. 25

Answer. Heat a haloalkane with aqueous NaOH or KOH; OH^- substitutes for halide to give the alcohol, $R–X+OH^- \rightarrow R–OH+X^-$.

Question 436 [H] · Original PDF p. 25

Answer. Aldehydes reduce to primary alcohols and ketones to secondary alcohols, e.g. with $NaBH_4$ or $LiAlH_4$ followed by work-up: $RCHO \rightarrow RCH_2OH$; $R_2CO \rightarrow R_2CHOH$.

Question 437 [M] · Original PDF p. 25

Answer. A primary alcohol under controlled oxidation gives an aldehyde, $RCH_2OH+[O] \rightarrow RCHO+H_2O$; distil promptly or use a mild reagent to limit further oxidation to the carboxylic acid.

Question 438 [H] · Original PDF p. 25

Answer. Primary alcohols oxidise first to aldehydes and, under stronger/continued conditions, carboxylic acids. Secondary alcohols give ketones. Tertiary alcohols resist mild oxidation; strong conditions may cleave C–C bonds.

Question 439 [M] · Original PDF p. 25

Answer. Lucas test uses concentrated HCl with anhydrous $ZnCl_2$ to convert alcohols to insoluble alkyl chlorides. Turbidity time distinguishes primary, secondary and tertiary alcohols approximately.

Question 440 [H] · Original PDF p. 25

Answer. Tertiary alcohols turn cloudy almost immediately, secondary alcohols after minutes, while primary alcohols show little/no turbidity at room temperature; carbocation stability controls the usual SN1-like rate order 3°>2°>1°.

Question 441 [M] · Original PDF p. 25

Answer. Ethanol forms sodium ethoxide and hydrogen: $2C_2H_5OH+2Na \rightarrow 2C_2H_5ONa+H_2\uparrow$.

Question 442 [M] · Original PDF p. 25

Answer. Dehydration removes H_2O from an alcohol, usually forming an alkene under acid/heat or an ether under suitable lower-temperature conditions.

Question 443 [H] · Original PDF p. 25

Answer. First protonate ethanol's OH to make H_2O a leaving group. Loss of water and removal of a β-H occur in an E2-like elimination for primary ethanol, forming $CH_2{=}CH_2$; acid is regenerated. No stable primary carbocation is required.

Question 444 [M] · Original PDF p. 25

Answer. Concentrated H_2SO_4 protonates –OH so it can leave as water and, as a dehydrating agent, favours water removal; temperature helps control alkene versus ether formation.

Question 445 [H] · Original PDF p. 25

Answer. Tertiary alcohols can form more stable carbocations after protonated water leaves, facilitating E1 dehydration; secondary are intermediate and primary are least favourable (often eliminate by a concerted route).

Question 446 [M] · Original PDF p. 25

Answer. Diethyl ether (ethoxyethane), $C_2H_5OC_2H_5$, forms predominantly at about 413 K: $2C_2H_5OH \rightarrow C_2H_5OC_2H_5 + H_2O$.

Question 447 [H] · Original PDF p. 25

Answer. At about 413 K, intermolecular dehydration gives diethyl ether. At about 443 K with concentrated H_2SO_4, intramolecular elimination gives ethene, $CH_2{=}CH_2$.

Question 448 [M] · Original PDF p. 25

Answer. Phenol is hydroxybenzene, with an –OH group directly attached to a benzene ring; formula C_6H_5OH (C_6H_6O).

Question 449 [M] · Original PDF p. 25

Answer. Phenol loses H^+ to form phenoxide, whose negative charge is delocalised into the aromatic ring. Ethoxide has no comparable resonance stabilisation, so ethanol is much less acidic.

Question 450 [H] · Original PDF p. 26

Answer. Phenoxide's negative charge is delocalised over oxygen and ortho/para ring positions by resonance. Ethoxide's charge remains localised mainly on oxygen, so phenoxide is more stable.

Question 451 [M] · Original PDF p. 26

Answer. An electron-withdrawing group stabilises phenoxide by inductive and/or resonance effects, especially at ortho/para positions, and increases phenol acidity.

Question 452 [H] · Original PDF p. 26

Answer. Acidity: p-nitrophenol>phenol>p-cresol. The para nitro group withdraws and stabilises phenoxide; the methyl group donates electron density and destabilises it.

Question 453 [M] · Original PDF p. 26

Answer. In the Dow process, chlorobenzene reacts with aqueous NaOH at high temperature and pressure to form sodium phenoxide; acidification gives phenol: $C_6H_5Cl \rightarrow C_6H_5ONa \rightarrow C_6H_5OH$.

Question 454 [M] · Original PDF p. 26

Answer. Bromine water rapidly substitutes at the activated ortho and para positions to give 2,4,6-tribromophenol as a white precipitate, decolourising bromine water.

Question 455 [H] · Original PDF p. 26

Answer. The –OH group donates a lone pair into the aromatic ring by resonance, increasing ring electron density and stabilising ortho/para arenium-ion intermediates; phenol reacts faster than benzene.

Question 456 [M] · Original PDF p. 26

Answer. Kolbe–Schmitt reaction: sodium phenoxide reacts with CO_2 under pressure/heat to form sodium salicylate; acidification yields salicylic acid (2-hydroxybenzoic acid).

Question 457 [M] · Original PDF p. 26

Answer. Reimer–Tiemann reaction formylates phenol mainly at the ortho position using $CHCl_3$ and aqueous NaOH, followed by acid work-up, to give salicylaldehyde.

Question 458 [H] · Original PDF p. 26

Answer. Main product is 2-hydroxybenzaldehyde (salicylaldehyde). $CHCl_3$/NaOH generates dichlorocarbene, $:CCl_2$, which introduces the formyl group after hydrolysis.

Question 459 [M] · Original PDF p. 26

Answer. Dilute HNO_3 gives a mixture of 2-nitrophenol and 4-nitrophenol by electrophilic substitution; the para isomer is often major.

Question 460 [H] · Original PDF p. 26

Answer. Dilute HNO_3 gives mainly ortho- and para-nitrophenol. Concentrated HNO_3 causes further nitration to 2,4,6-trinitrophenol (picric acid).

Question 461 [M] · Original PDF p. 26

Answer. Add neutral $FeCl_3$: phenol gives a violet/purple coloured complex, whereas ethanol does not give this characteristic colour.

Question 462 [M] · Original PDF p. 26

Answer. An ether has an oxygen atom bonded to two carbon groups: R–O–R′ (which may be identical or different).

Question 463 [M] · Original PDF p. 26

Answer. Methoxyethane.

Question 464 [M] · Original PDF p. 26

Answer. Williamson synthesis: react a sodium alkoxide/phenoxide with a suitable primary alkyl halide by SN2: $RONa + R'X \rightarrow ROR' + NaX$.

Question 465 [H] · Original PDF p. 26

Answer. A primary halide is least sterically hindered and undergoes SN2 substitution readily. Secondary/tertiary halides increasingly favour E2 elimination with the alkoxide base.

Question 466 [H] · Original PDF p. 26

Answer. Propene is usually the major product by E2 elimination because 2-bromopropane is secondary and ethoxide is a strong base. A smaller amount of ethyl isopropyl ether may form by competing SN2 substitution.

Question 467 [M] · Original PDF p. 26

Answer. Excess hot HI cleaves both C–O bonds of a dialkyl ether: $C_2H_5OC_2H_5 + 2HI \rightarrow 2C_2H_5I + H_2O$.

Question 468 [H] · Original PDF p. 26

Answer. Protonate ether oxygen, then I^- usually attacks the less hindered alkyl group by SN2. If one side is tertiary, cleavage at that side can proceed by SN1; in aryl–alkyl ethers the alkyl–O bond breaks, giving phenol and alkyl iodide.

Question 469 [M] · Original PDF p. 26

Answer. Alcohols can donate and accept intermolecular hydrogen bonds; ethers can accept but cannot donate O–H hydrogen bonds. Weaker intermolecular association gives ethers lower boiling points than isomeric alcohols.

Question 470 [M] · Original PDF p. 26

Answer. Diethyl ether has been used as a solvent and historically as a general anaesthetic; its flammability and peroxide hazard require careful handling.

Question 471 [H] · Original PDF p. 26

Answer. Oxygen and light can initiate radical autoxidation at α-C–H positions of ethers, forming explosive hydroperoxides/peroxides that may concentrate during storage or distillation.

Question 472 [M] · Original PDF p. 26

Answer. Methanol is metabolised to formaldehyde and formate, causing metabolic acidosis, optic-nerve damage/blindness and potentially death; even small ingestions are medical emergencies.

Question 473 [M] · Original PDF p. 26

Answer. Ethanol dissolves many polar and moderately non-polar substances, mixes with water, evaporates readily and is relatively accessible, making it useful in formulations, cleaning and extraction.

Question 474 [H] · Original PDF p. 27

Answer. Methanol is made mainly from synthesis gas and used as a feedstock/solvent/fuel; it is highly toxic and can cause blindness or death. Ethanol is made by fermentation or ethene hydration and used as solvent/fuel; drinking it is intoxicating and excessive exposure is harmful.

Question 475 [M] · Original PDF p. 27

Answer. Esterification is formation of an ester from an alcohol and a carboxylic acid, usually reversibly under acid catalysis: $RCOOH + R'OH \rightleftharpoons RCOOR' + H_2O$.

Question 476 [H] · Original PDF p. 27

Answer. $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$. Acid catalyses carbonyl activation and proton transfers; it accelerates attainment of equilibrium but does not change the equilibrium constant.

Question 477 [M] · Original PDF p. 27

Answer. Water solubility generally decreases as the carbon chain length increases: the non-polar hydrophobic portion becomes larger relative to the hydrogen-bonding –OH group.

Question 478 [H] · Original PDF p. 27

Answer. Phenol's –OH can hydrogen-bond with water, but its large hydrophobic phenyl ring is poorly hydrated; this limits solubility compared with lower alcohols.

Question 479 [M] · Original PDF p. 27

Answer. On oxidation, primary alcohols give aldehydes and then carboxylic acids; secondary alcohols give ketones; tertiary alcohols resist ordinary mild oxidation and require harsh conditions that may cleave C–C bonds.

Question 480 [M] · Original PDF p. 27

Answer. 2-Methylpropan-2-ol.

Question 481 [H] · Original PDF p. 27

Answer. 2-Methylpropene, $(CH_3)_2C{=}CH_2$, is the major alkene from acid-catalysed dehydration of 2-methylpropan-2-ol.

Question 482 [M] · Original PDF p. 27

Answer. Phenol ($pK_a \approx 10$) is acidic enough for OH^- to deprotonate it, forming resonance-stabilised phenoxide and water. Ethanol ($pK_a \approx 16$) is much weaker; OH^- does not appreciably deprotonate it in water.

Question 483 [H] · Original PDF p. 27

Answer. Electron-withdrawing −I groups stabilise phenoxide and increase acidity; at ortho/para positions, −M groups can further delocalise charge by resonance. Electron-donating +I/+M groups destabilise phenoxide and reduce acidity; effects depend on position.

Question 484 [M] · Original PDF p. 27

Answer. Controlled bromination in a non-polar solvent (e.g. Br_2 in CS_2, low temperature) gives mainly p-bromophenol, with some o-bromophenol; conditions limit further substitution.

Question 485 [H] · Original PDF p. 27

Answer. Bromine water gives 2,4,6-tribromophenol rapidly because phenoxide-like resonance strongly activates all ortho/para positions. In a non-polar solvent under controlled conditions, monobromination occurs, with para usually favoured over ortho.

Question 486 [M] · Original PDF p. 27

Answer. React sodium phenoxide with a primary methyl halide, e.g. $C_6H_5ONa + CH_3I \rightarrow C_6H_5OCH_3 + NaI$ (anisole), by SN2 at methyl.

Question 487 [H] · Original PDF p. 27

Answer. Anisole + HI gives phenol and CH_3I: I^- attacks the methyl group by SN2 after oxygen protonation. The aryl C(sp^2)–O bond is not cleaved by ordinary SN2, even with excess HI.

Question 488 [M] · Original PDF p. 27

Answer. A phenol oxygen lone pair is conjugated with the aromatic π system, giving the aryl C–O bond partial double-bond character; this makes it shorter than a typical alcohol C–O single bond.

Question 489 [H] · Original PDF p. 27

Answer. Acid protonates OH, allowing water to leave. Tertiary carbocations are most stable, then secondary, then primary, so ionisation/E1 dehydration is generally 3°>2°>1°; primary alcohols often use concerted elimination instead.

Question 490 [M] · Original PDF p. 27

Answer. Propan-2-ol oxidises to propanone (acetone), CH_3COCH_3.

Question 491 [H] · Original PDF p. 27

Answer. Oxidise under controlled conditions: 1-propanol gives propanal (and with stronger/continued oxidation, propanoic acid); 2-propanol gives propanone. The aldehyde/acid versus ketone products distinguish them.

Question 492 [M] · Original PDF p. 27

Answer. Acid protonates/activates the carbonyl and facilitates nucleophilic addition and proton transfers, speeding esterification. It is regenerated and does not alter the equilibrium position.

Question 493 [H] · Original PDF p. 28

Answer. Treat phenol with NaOH to form sodium phenoxide; react with CO_2 under pressure/heat to form sodium salicylate; acidify to obtain salicylic acid. This is the Kolbe–Schmitt reaction.

Question 494 [M] · Original PDF p. 28

Answer. Phenol gives a violet/purple coloured ferric–phenoxide complex with neutral $FeCl_3$.

Question 495 [H] · Original PDF p. 28

Answer. The colour is not uniquely conclusive: enols and some other compounds can also complex Fe^{3+}, while pH, concentration, oxidation and sample colour affect observations. Confirm with an independent test or spectroscopy.

Question 496 [M] · Original PDF p. 28

Answer. Lower alcohols have a relatively small hydrophobic alkyl group and an –OH group that hydrogen-bonds strongly with water, making mixing favourable and often miscible.

Question 497 [H] · Original PDF p. 28

Answer. Branching makes alcohol isomers more compact, reducing contact area and dispersion forces; boiling points generally fall with branching. Hydrogen bonding remains, so exact trends can show exceptions.

Question 498 [M] · Original PDF p. 28

Answer. Excess hot HI cleaves both ether bonds: $(C_2H_5)_2O + 2HI \rightarrow 2C_2H_5I + H_2O$; ethyl iodide is the major organic product.

Question 499 [H] · Original PDF p. 28

Answer. Ethers have no acidic proton, so ordinary bases do not readily break their C–O bonds. Strong HI/HBr protonates oxygen; the resulting oxonium ion is attacked by halide, cleaving a C–O bond.

Question 500 [H] · Original PDF p. 28

Answer. NaOH deprotonates phenol to sodium phenoxide, C_6H_5ONa. CO_2 under pressure followed by acidification gives salicylic acid (2-hydroxybenzoic acid); this is the Kolbe–Schmitt reaction.

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